

COMPUTER NETWORKS

Sessional Exam Study Notes

6CS105CC26 · MCA Semester I · Nirma University

Unit 1: Introduction and Network Types · Unit 2: Framing and ARQ · Unit 3: Data and Signals · Unit 4: Switching

Built from your syllabus slides, 4 past papers and the Question Bank. Every topic is priority-tagged (🔥 / ★ / 📐), explained with an analogy, a worked example and a figure.

How to Use These Notes

1. **Look at the tag first.** Every topic heading carries a priority tag taken from the past-paper analysis. Spend most of your time on the 🔥 topics.
2. **Read the analogy before the theory.** If you can explain the analogy to a friend, the theory is already half-learned.
3. **Redraw every figure by hand.** Examiners award marks for neat sketches (the papers say "draw neat sketches wherever necessary"). The ASCII figures here are designed to be easy to redraw.
4. **Solve every worked example again with new numbers.** Past papers reuse the same question template with changed values.
5. **Finish with the Practice Questions** at the end, then check yourself against the Model Answer Skeletons.

Priority Tag Legend

Tag	Meaning	How to study it
[🔥 HIGH-YIELD: Frequently Asked]	Appeared repeatedly in past papers and/or has several Question Bank items. Likely to be a full 6–8 mark question.	Learn completely: theory, diagram, example, comparison. Practise writing it in full.
[★ ESSENTIAL FOR EXAM]	Listed in the Question Bank or partly asked in past papers. Easy-to-medium marks, often numericals or definitions.	Learn the definition, one diagram and the formulas. Practise the numericals.
[📐 SUPPORTING TOPIC]	Not asked directly so far, but it is in your slides and builds the foundation for the other topics.	Read once for understanding; memorise the one-line definitions for 1–3 mark questions.

Priority Map of All Sessional Topics

Section	Topic	Tag	Past-paper evidence
1.1	Core terms (network, topology, unicast/multicast/broadcast, analog/digital, bandwidth)	★ ESSENTIAL	Jan 2023 Q2A; QB Q1(a–m)
1.2	Data communication vs computer network	📐 SUPPORTING	QB Q1(a, b)
1.3 – 1.6	Components, data representation, protocols, data flow	📐 SUPPORTING	Slides only

Section	Topic	Tag	Past-paper evidence
1.7	Point-to-point and multipoint connections	★ ESSENTIAL	QB Q1(c, d)
1.8	Network topologies (Mesh, Star, Bus, Ring, Hybrid, Tree)	🔥 HIGH-YIELD	Jan 2023 Q3A; Jan 2025 Q1b; QB Q2–5
1.9	PAN, LAN, MAN, WAN and wireless networks	🔥 HIGH-YIELD	Jan 2022 Q1; QB Q6–9
1.10	Switching overview and the Internet	📘 SUPPORTING	Slides only
2.1	Role of the Data Link Layer	★ ESSENTIAL	Dec 2025 Q1a (partly)
2.2	Framing: byte stuffing and bit stuffing	🔥 HIGH-YIELD	Jan 2023 Q3B; Dec 2025 Q2b; QB Q30–32
2.3	Flow control and error control	★ ESSENTIAL	Dec 2025 Q1a (partly)
2.4	Noiseless-channel protocols (Simplest, Stop-and-Wait)	★ ESSENTIAL	Base for ARQ questions
2.5	Stop-and-Wait ARQ	🔥 HIGH-YIELD	QB Q33, Q36; baseline in SR-vs-GBN
2.6	Pipelining and sliding window	★ ESSENTIAL	QB Q38–39 numericals
2.7	Go-Back-N ARQ	🔥 HIGH-YIELD	Jan 2023 Q1B; Jan 2025 Q2c; QB Q34, Q37
2.8	Selective Repeat ARQ	🔥 HIGH-YIELD	Jan 2022 Q4; Jan 2023 Q1B; Jan 2025 Q2c; QB Q35
2.9	Selective Repeat vs Go-Back-N (full comparison)	🔥 HIGH-YIELD (#1)	3 of 4 papers
2.10	Window-size and utilisation numericals	★ ESSENTIAL	QB Q38, Q39
3.1 – 3.3	Analog vs digital, sine-wave parameters, time vs frequency domain	📘 SUPPORTING	QB Q1(i–l)
3.4	Composite signals	📘 SUPPORTING	Foundation for bandwidth
3.5	Bandwidth and spectrum numericals	🔥 HIGH-YIELD	Jan 2022 Q8; QB Q19, Q20
3.6	Digital signals: bit rate, bit length, baseband	★ ESSENTIAL	QB Q21, Q22
3.7	Transmission impairment: attenuation, distortion, noise, dB, SNR	★ ESSENTIAL	QB Q23
4.1 – 4.2	Why switching; circuit switching	★ ESSENTIAL	QB Q55, Q56
4.3 – 4.5	Datagram (packet), virtual-circuit and message switching	★ ESSENTIAL	QB Q55, Q57, Q58
4.6	Comparison of all switching techniques	★ ESSENTIAL	QB Q55

Suggested 5-Session Study Plan

Session	Cover	Goal at the end
1	Unit 1: 1.8 Topologies, 1.9 PAN/LAN/MAN/WAN, 1.1 Core terms	Draw all 4 topologies and write the mesh formula without looking.
2	Unit 2: 2.1 – 2.3 Data link layer, Framing	Stuff and unstuff a bit string and a byte string on paper.
3	Unit 2: 2.4 – 2.10 ARQ family	Draw the Stop-and-Wait, GBN and SR lost-frame timelines from memory; recite the SR vs GBN table.

Session	Cover	Goal at the end
4	Unit 3: bandwidth, bit rate, impairment	Solve all numericals in 3.5, 3.6 and 3.7 in under 3 minutes each.
5	Unit 4 + Practice Questions	Fill the 4-way switching comparison table from memory; attempt 10 practice questions timed.

⚠ Scope Note (read once)

These notes cover only the sessional slides you uploaded: Introduction (slides 1–58), Data Link Layer framing and ARQ (Chapter 11), Data and Signals (Chapter 3) and Switching (Chapter 8).

Selective Repeat ARQ is only mentioned on three slides in your deck, so Section 2.8 is built from the standard textbook treatment. **Unicast and multicast** are also not explained in the slides; Section 1.1 covers them because they are asked in past papers.

Layered models (OSI/TCP-IP), error-detection codes (CRC, Hamming) and MAC protocols (ALOHA, CSMA) are not in your sessional slides, so they are not included. If your faculty added them, tell your tutor and they can be added.

UNIT 1 — Introduction to Data Communication and Networks

What this unit is about: how devices talk to each other, what a network is made of, how networks are shaped (topologies) and how they are classified by size (PAN, LAN, MAN, WAN). Slides 1–58.

1.1 Core Terms You Must Be Able to Define [★ ESSENTIAL FOR EXAM]

Exam Insight

Asked as: Jan 2023 Q2A (distinguish unicast, multicast, broadcast, 6 marks). Question Bank Q1(a–m) lists 13 short definitions worth 1–3 marks each.

Why it matters: these are the cheapest marks in the paper. A crisp one-line definition plus a one-word example earns full marks.

Simple Analogy

Think of a **school**. The school (a **network**) connects students, teachers and offices. Teachers speaking to a class is **communication**. The way classrooms are arranged in the corridor is the **topology**. How wide the corridor is decides how many people can walk at once: that is **bandwidth**.

Term	Exam-ready definition	Real-world example
Computer network	A collection of interconnected computers and other devices that can communicate with each other and share resources.	A university LAN connecting hundreds of PCs, printers and servers.
Data communication	The exchange of data between two devices through some transmission medium (wired or wireless), using hardware and software.	Sending a file from PC A to PC B.
Point-to-point connection	A dedicated link between exactly two devices; the entire capacity of the link is reserved for those two devices.	A leased line between two offices.
Multipoint connection	More than two devices share the same link, so the channel capacity is shared.	A wireless access point serving many phones.
Network topology	The geometric arrangement of links and nodes, showing how all devices are connected to one another.	Star layout of a computer lab with one switch in the centre.
Unicast	One sender to exactly one receiver.	A private WhatsApp message.
Multicast	One sender to a selected group of receivers.	A video call with four chosen participants; a YouTube live stream to subscribers.
Broadcast	One sender to all devices on the network.	A college PA announcement heard in every room; TV broadcast.
Analog data	Information that is continuous and takes continuous values.	Human voice, temperature reading from a mercury thermometer.
Analog signal	A signal that can take an infinite number of values in a range.	The smooth wave from a microphone.
Digital data	Information with discrete states that takes discrete values.	Text stored as 0s and 1s.
Digital signal	A signal that can have only a limited number of values.	Voltage switching between 0 V and 5 V.

Term	Exam-ready definition	Real-world example
Bandwidth	For a signal: the difference between the highest and lowest frequencies it contains. For a link: the maximum data rate it can carry.	A 200 kHz FM radio channel; a 100 Mbps broadband plan.

Unicast vs Multicast vs Broadcast (6-mark answer)

 Three delivery modes		
UNICAST (1 -> 1)	MULTICAST (1 -> group)	BROADCAST (1 -> all)
[S] -----> [R]	[S] ----+----> [R1] +----> [R2] (R3 not in group)	[S] ----+----> [R1] +----> [R2] +----> [R3] +----> [R4]

Feature	Unicast	Multicast	Broadcast
Receivers	Exactly one	A chosen group	Every device in the network
Bandwidth use	One copy per receiver (heavy if many)	One copy shared by the group	One copy; reaches everyone, even those who do not need it
Privacy	High	Medium	None
Example	Email to one friend	Live class video to enrolled students	Wi-Fi beacon, TV channel

Exam Tip

In a 1–3 mark "define" question, write: **definition + one example**, nothing more. Two lines is enough. For the 6-mark "distinguish" question, draw the three small diagrams above and add the comparison table.

30-Second Recap

Network = connected devices sharing resources. Data communication = the act of exchanging data.
 Point-to-point = 2 devices, full capacity. Multipoint = many devices, shared capacity.
 Unicast = 1 to 1, multicast = 1 to some, broadcast = 1 to all.
 Bandwidth (signal) = $f_{\text{high}} - f_{\text{low}}$.

1.2 Data Communication vs Computer Network SUPPORTING TOPIC: Learn for short answers]

Exam Insight

Asked as: QB Q1(a, b) definitions. Slide 7 gives a five-row comparison; a table like the one below is the ideal answer.

Simple Analogy

Data communication is the traffic; the computer network is the road system. Without roads (network) there is nowhere for traffic to go, and roads with no traffic are pointless. Data communication is a function of a network.

Aspect	Data communication	Computer network
Meaning	Exchange of data between devices	Interconnection of devices

Aspect	Data communication	Computer network
Main focus	Transmission of data	Connectivity and resource sharing
Components	Sender, receiver, message, medium, protocol	Computers, switches, routers, cables, protocols
Scope	Can occur between just two devices	Usually many interconnected devices
Example	Sending a file from PC A to PC B	A university LAN of hundreds of computers

Four characteristics of a good data communication system

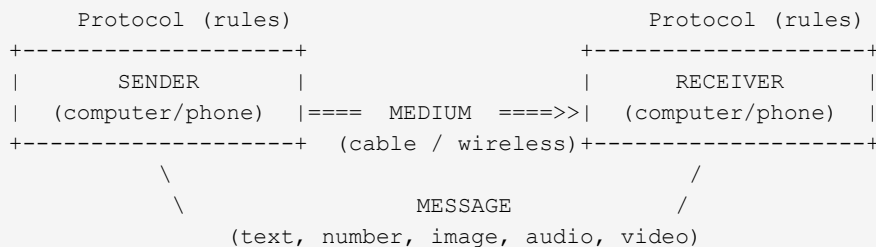
Characteristic	Meaning	Example
Delivery	Data must reach the correct destination.	An email goes to the right inbox.
Accuracy	Data must arrive without corruption.	A downloaded PDF opens without errors.
Timeliness	Data must arrive in time to be useful.	A live cricket stream should not lag by minutes.
Jitter	Variation in the arrival time of packets (uneven delay).	Robotic, broken voice on a video call even though no packet is lost.

1.3 The Five Components of Data Communication [■ SUPPORTING TOPIC: Learn for short answers]

Simple Analogy

Sending a **letter**: you (sender) write a letter (message), the postal system (medium) carries it, the postal rules (protocol) say how to address it, and your friend (receiver) reads it.

Five components: message, sender, receiver, medium, protocol



- **Message:** the information to be communicated (text, numbers, images, audio, video).
- **Sender:** the device that sends the message (computer, phone, camera).
- **Receiver:** the device that receives the message.
- **Transmission medium:** the physical path (twisted-pair cable, fibre, radio waves).
- **Protocol:** the set of rules both sides agree on. Without it, two devices are like two people speaking different languages.

1.4 Data Representation [■ SUPPORTING TOPIC: Learn for short answers]

Computers store everything as bits (0s and 1s). Each type of information has its own coding scheme.

Type	How it is represented	Example
Text	Each symbol is a bit pattern. The set of patterns is called a code (ASCII, Unicode).	The letter A = 65 = 01000001 in ASCII.

Type	How it is represented	Example
Numbers	Converted directly into binary bit patterns.	Decimal 13 = 1101.
Images	A matrix of pixels; each pixel has a bit pattern. Black-and-white needs 1 bit per pixel; coloured images use RGB, CMY or HSV.	A 1-bit pixel: 0 = black, 1 = white.
Audio	A continuous recording of sound that is sampled and converted to bits.	A voice call.
Video	A continuous entity or a sequence of images shown rapidly.	A movie file.

Code snippet: character to bit pattern

```
# Python: see how text becomes bits
for ch in "CN":
    print(ch, ord(ch), format(ord(ch), "08b"))

# Output:
# C 67 01000011
# N 78 01001110
```

1.5 Network Communication Protocols [SUPPORTING TOPIC: Learn for short answers]

Definition (write this in the exam)

A **protocol** is a set of rules that governs data communication. It defines **what** is communicated, **how** it is communicated and **when** it is communicated. Its three key elements are syntax, semantics and timing.

Element	Meaning	Example
Syntax	The structure or format of the data.	A frame must be: 8-bit sender address, then 8-bit receiver address, then data.
Semantics	The meaning of each section of bits.	The first 8 bits mean "who sent this"; the next 8 bits mean "who should receive it".
Timing	When data should be sent and how fast.	If a sender produces data at 100 Mbps but the receiver can only process 1 Mbps, the receiver is overloaded and data is lost.

Simple Analogy

A **phone call**. Syntax = you say "Hello" first, then speak. Semantics = "Hello" means "I am ready to talk". Timing = you do not talk so fast that the other person cannot follow.

1.6 Data Flow: Simplex, Half-Duplex, Full-Duplex [SUPPORTING TOPIC: Learn for short answers]

Exam Insight

Not asked directly in past papers, but it is the classic "types of transmission mode" answer and appears in slides 14–18.

 Three modes of data flow**SIMPLEX**

[A] -----> [B]
one way only

HALF-DUPLEX

[A] <-----> [B]
both ways, but ONE
direction at a time

FULL-DUPLEX

[A] <=====> [B]
both ways at the SAME
time

Mode	How it works	Real-world examples	Analogy
Simplex	Data travels in only one direction. The receiver cannot send back on the same channel.	TV and radio broadcast, keyboard to computer, computer to printer.	A one-way street.
Half-duplex	Both devices can send and receive, but not at the same time.	Walkie-talkie, police radio, old Ethernet hubs.	A single-lane bridge where cars take turns.
Full-duplex	Both devices can send and receive simultaneously.	Phone call, video conferencing, modern switched Ethernet.	A two-lane road.

1.7 Network Criteria and Physical Structures [★ ESSENTIAL FOR EXAM]

 Exam Insight

Asked as: QB Q1(c, d) — define point-to-point and multipoint connection. Network criteria (performance, reliability, security) are slide 20.

Point-to-point vs multipoint

 Two types of connection**POINT-TO-POINT**

[A] ===== [B]
dedicated link, full capacity

MULTIPOINT

[A] [B] [C] [D]
| | | |
---+---+---+---+--- (shared link)

Feature	Point-to-point	Multipoint
Devices on the link	Exactly two	More than two
Capacity	Entirely reserved for the two devices	Shared among all devices
Examples	Leased line between two offices; direct fibre between two routers	Wireless access point serving many clients; satellite uplink to many ground stations

Network criteria

- **Performance:** measured by transit time, response time, throughput and delay. Affected by number of users, medium type, hardware and software efficiency.
- **Reliability:** how often the network fails, how fast it recovers, and how robust it is in a disaster.
- **Security:** protecting data from unauthorised access and damage, and having recovery policies for breaches and data loss.

1.8 Network Topologies [🔥 HIGH-YIELD: Frequently Asked]

📖 Exam Insight

Asked as: Jan 2023 Q3A (cables and ports for 5 devices in mesh, 4 marks); Jan 2025 Q1b (differentiate Star, Bus, Mesh, Ring with diagrams, 8 marks). Question Bank Q2–5 ask for a full "explain in detail with advantages and disadvantages" answer for each topology, 5–8 marks each.

Frequency: 2 of 4 past papers + 4 QB questions. Treat this as a must-know.

🧠 Simple Analogy

A topology is the **floor plan of a network**, like the layout of roads in a town.

- **Mesh** = every house has its own private road to every other house (very safe, very expensive).
- **Star** = every house connects to one central roundabout.
- **Bus** = all houses along one long main road.
- **Ring** = houses arranged around a circular road.

📝 How to answer any topology question (5 steps)

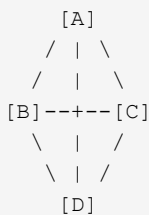
1. Write a **one-line definition** of how devices are connected.
2. Draw a **neat labelled diagram**.
3. Explain **how data travels**.
4. List **4–5 advantages** and **4–5 disadvantages**.
5. Add a **real-world example**.

A. Mesh Topology

📖 Definition (write this in the exam)

In a mesh topology every device has a **dedicated point-to-point link to every other device**.

🔍 Mesh topology (4 nodes)



Full mesh of 4 nodes (A, B, C, D)
Every pair has its own link:
A-B, A-C, A-D, B-C, B-D, C-D = 6 links

Counting links and ports (numerical)

Quantity	Formula	Reason
Physical links, full-duplex	$n(n - 1) / 2$	Each of n nodes connects to the other $n-1$ nodes; every link is counted twice, so divide by 2.
Physical links, simplex	$n(n - 1)$	Two links per pair (one for each direction).
Ports needed on each device	$n - 1$	One port for every other device.
Total ports in the network	$n \times (n - 1)$	n devices $\times$ $(n-1)$ ports each.

🔧 Worked Example (Jan 2023 Q3A): 5 devices in a mesh

Given $n = 5$.

- Cables (full-duplex) = $n(n-1)/2 = 5 \times 4 / 2 = \mathbf{10 \text{ cables}}$.
- Ports per device = $n - 1 = \mathbf{4 \text{ ports}}$.
- Total ports = $5 \times 4 = \mathbf{20 \text{ ports}}$.

Check by listing: with devices A–E the pairs are AB, AC, AD, AE, BC, BD, BE, CD, CE, DE = 10 pairs. ✓

Try yourself: 10 devices $\rightarrow 10 \times 9 / 2 = 45$ cables, 9 ports per device.

📄 Code snippet: mesh formula

```
# Python: links in a full mesh
def mesh(n, duplex=True):
    links = n * (n - 1)
    return links // 2 if duplex else links

print(mesh(5))           # 10
print(mesh(5, False))    # 20
print(mesh(10))          # 45
```

Advantages	Disadvantages
• High reliability: if one link fails, data takes another path. • Fault tolerance: one failure rarely brings down the network. • Dedicated links: no sharing, so no congestion. • Better security and privacy: traffic is isolated. • Easy fault identification: each link is separate. • Suitable for critical applications: backbones, telecom.	• High installation cost: a very large number of cables. • Complex installation as devices increase. • Difficult maintenance and troubleshooting. • Needs many ports on every device. • Not easily scalable: adding a device needs n new links. • High management complexity.

🌐 Real-World Example

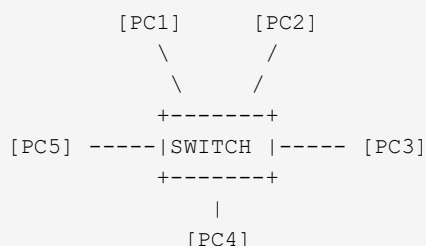
Regional telephone offices connected to each other, and WAN backbones. Where downtime is unacceptable, the high cost is justified.

B. Star Topology

📖 Definition (write this in the exam)

In a star topology every device has a **dedicated point-to-point link only to a central controller** (hub or switch). Devices do not communicate directly; the controller relays all traffic.

🔗 Star topology



Every PC has ONE cable to the switch.
PC1 $\rightarrow$ PC3 goes PC1 $\rightarrow$ Switch $\rightarrow$ PC3

Advantages	Disadvantages
• Easy to install and manage. • Easy fault detection: a faulty cable affects only one device. • One device failing does not affect the others. • Easy to expand: plug into a free port. • Better performance with a switch (dedicated connections). • Good security: switch can enforce port security.	• Central device is a single point of failure. • Higher cabling cost: one cable per device. • Depends heavily on the central device. • Limited by number of ports. • Central device can become a bottleneck. • Hard to wire in large buildings.

Real-World Example

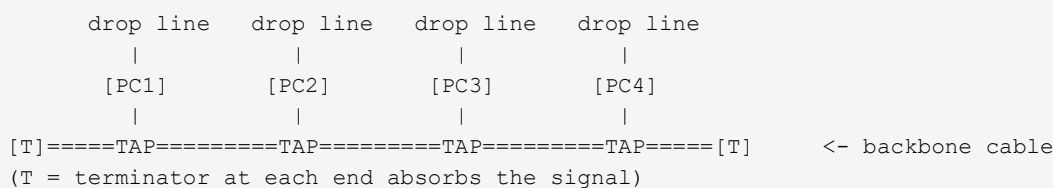
A computer lab where every PC is plugged into one switch; most home and office LANs.

C. Bus Topology

Definition (write this in the exam)

A bus topology is a **multipoint** topology in which **one long cable acts as a backbone** that links all devices. Nodes connect to it through **drop lines and taps** (a tap is a connector that splices into the main cable).

Bus topology



Advantages	Disadvantages
• Easy to install (one main cable). • Low cost: less cable than star or mesh. • No central device needed. • Easy to extend in small networks. • Good for small, low-traffic networks. • Simple structure.	• Backbone failure brings down the whole network. • Hard to troubleshoot: all devices share one cable. • Performance drops as more devices join. • Collisions when two devices send together (old Ethernet used CSMA/CD). • Limited cable length and limited number of devices. • Weak security: everyone sees the shared medium.

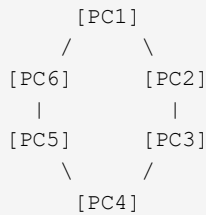
Real-World Example

Early Ethernet (10BASE2/10BASE5) coaxial networks in small labs.

D. Ring Topology

Definition (write this in the exam)

In a ring topology each device has a **dedicated point-to-point connection with only the two devices on either side of it**. A signal travels around the ring in one direction; each device's **repeater regenerates the bits** and passes them on.

 Ring topology (6 devices)

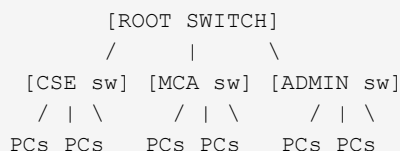
Data travels one way:

PC1 -> PC2 -> PC3 -> ... -> PC6 -> PC1

Advantages	Disadvantages
• Fewer collisions: token passing controls who may transmit. • Efficient transmission along a predefined path. • Equal access: every device gets a turn. • Good under heavy traffic. • No central device needed. • Direction of data flow is predictable.	• One broken device or link can break the ring. • Hard to troubleshoot a circular arrangement. • Hard to add or remove devices without interrupting the network. • Expansion is difficult. • Delay: a packet may pass through many devices. • Maintenance is more complex than star.

 Real-World Example

IBM Token Ring LANs (IEEE 802.5, 4–16 Mbps) and fibre-based ring networks such as FDDI.

E. Hybrid and Tree Topologies (bonus)**Hybrid:** a combination of two or more topologies. For example, a main star with each branch connecting several stations in a bus.**Tree (hierarchical):** a collection of star networks arranged in a hierarchy, like branches of a tree. Highly scalable and easy to manage; but a failure at the root or backbone can cut off many branches. **Tree topology**

A university tree network:

root = campus core,

branches = departments,

leaves = PCs in each lab.

Master comparison table (use this for the 8-mark "differentiate" question)

Feature	Mesh	Star	Bus	Ring
Connection	Link to every other device	Link to central hub/switch	All on one backbone	Link to two neighbours
Cabling	Most $(n(n-1)/2)$	One cable per device	Least	One cable per device
Cost	Highest	Medium-high	Lowest	Medium
Reliability	Very high	Good (except centre)	Poor	Poor (single ring)
Single point of failure	None	Central switch	Backbone cable	Any node or link
Fault finding	Easy	Easy	Difficult	Difficult
Adding a device	Hard (n new links)	Easy (free port)	Easy (small nets)	Hard
Collisions	None	Few (with switch)	Common	Few (token)
Example	Telecom backbone	Office/lab LAN	Old coax Ethernet	Token Ring

⚠ Common Mistake

Writing "mesh = $n(n-1)/2$ " without saying **full-duplex**. Always state the mode: $n(n-1)/2$ for full-duplex, $n(n-1)$ for simplex.

Saying a star network "has no failure point". The central switch **is** the single point of failure.

Forgetting terminators and taps in the bus diagram, or arrows in the ring diagram.

☑ 30-Second Recap

Mesh = every-to-every, $n(n-1)/2$ links, very reliable, very costly. Star = all to a centre, easy but centre is critical.

Bus = one backbone, cheap but one cut kills it. Ring = neighbours only, data goes one way, one break hurts.

Hybrid = mix. Tree = stars arranged in a hierarchy.

1.9 Categories of Networks: PAN, LAN, MAN, WAN [🔥 HIGH-YIELD: Frequently Asked]

📖 Exam Insight

Asked as: Jan 2022 Q1 — "Explain the key characteristics of PAN, LAN, MAN and WAN" (8 marks, the opening question). Question Bank Q6 (PAN, 2–3 marks), Q7, Q8, Q9 (LAN, MAN, WAN, 5–8 marks each).

Best strategy: learn one comparison table plus one line of "key characteristics" for each network.

🧠 Simple Analogy

Imagine **who you can talk to by shouting**. PAN = a conversation at your desk. LAN = the whole office building. MAN = the whole city through a loudspeaker network. WAN = the entire world through the telephone system.

📊 Networks by size (smallest to largest)

PAN	LAN	MAN	WAN
(~1 room)	(building/campus)	(a city)	(country / continent / world)
<pre> . . . '-'-' 10-100 m </pre>	<pre> .----- LAN LAN LAN '----- up to few km </pre>	<pre> .----- LAN LAN LAN LAN LAN '----- city-wide </pre>	<pre> .----- MAN LAN MAN LAN (many cities) '----- very large area </pre>

A. Personal Area Network (PAN)

- A network meant for **one person**, covering about **10 to 100 metres** (up to one room).
- Connects personal devices: computer, phone, tablet, printer, headset, speakers, game console.
- Flexible and efficient for short ranges; often wireless using **Bluetooth**.

🌐 Real-World Example

Connecting your phone to a Bluetooth earbud, or a laptop to a nearby printer.

B. Local Area Network (LAN)

- A **privately owned** network in a single building or campus; covers up to a **few kilometres**.
- Used by businesses, schools and homes; typically fewer than a few hundred devices.
- **Very high speed:** traditionally 10–100 Mbps, today up to 10 Gbps.

- A LAN is characterised by three things: **size** (known in advance, so design is easy), **transmission technology** (cable or wireless) and **topology** (bus, ring, star).
- **Broadcast LANs**: only one machine transmits at a time; if two collide, each waits a random time and retries. Examples: **IEEE 802.3 Ethernet** (10 Mbps–10 Gbps) and **IEEE 802.5 Token Ring** (4–16 Mbps).
- **Static** broadcast LANs give each machine a turn in round-robin fashion (wastes capacity if a machine has nothing to send); **dynamic** ones allocate the channel on demand, with centralised or decentralised control.

Real-World Example

The Wi-Fi/Ethernet network of one college campus, or your home Wi-Fi.

C. Metropolitan Area Network (MAN)

- Covers an **entire city** or a large area; essentially a **collection of LANs** in the same geographic area.
- A high-speed network (not as fast as a LAN) for sharing regional resources.
- Typically uses wired infrastructure such as cables. A classic example is the **cable-television network**, which later became a two-way Internet service.
- High-speed wireless Internet access is also a MAN example, standardised as **IEEE 802.16**.

Real-World Example

A city-wide cable-TV and broadband network; a city's CCTV network linked to one control room.

D. Wide Area Network (WAN)

- Spans a **large geographic area**: a region, country, continent or the whole world, connecting many LANs and MANs.
- Links are provided by **public carriers** (telephone companies, ISPs, cable systems, satellites).
- Generally has a **higher bit error rate**, **lower data rate** and **higher propagation delay** than a LAN.
- The **communication subnet** has two parts: **transmission lines** (copper, fibre, radio) that move bits, and **switching elements** (routers) that decide the outgoing line for incoming data.
- If a router stores packets and sends them when the output line is free, the subnet is a **store-and-forward (packet-switched) subnet**.
- Satellite WANs are inherently **broadcast**.

Real-World Example

A bank's network linking its branches in Ahmedabad, Mumbai and Delhi; the Internet backbone between countries.

Master comparison table

Feature	PAN	LAN	MAN	WAN
Full form	Personal Area Network	Local Area Network	Metropolitan Area Network	Wide Area Network
Area covered	10–100 m (one room)	Building / campus, few km	A city	Country / continent / world
Ownership	One person	Private (one organisation)	Private or public	Public carriers / many organisations
Speed	Moderate	Very high (up to 10 Gbps)	High (below LAN)	Lower; high delay and error rate

Feature	PAN	LAN	MAN	WAN
Typical technology	Bluetooth, USB	Ethernet (802.3), Wi-Fi (802.11)	Cable TV, IEEE 802.16	Telephone lines, satellites, fibre
Example	Phone + earbuds	College campus	City cable broadband	The Internet

Wireless networks (slides 51–52)

Category	Description	Standard / example
System interconnection	Short-range radio linking computer, mouse, printer, headset, camera. No cables or driver installation; just switch on.	Bluetooth
Wireless LAN	Every computer has a radio modem and antenna; talks to a ceiling access point or directly peer-to-peer.	IEEE 802.11 (Wi-Fi)
Wireless WAN	Radio networks such as cellular telephone (low bandwidth) and high-speed wireless access.	Cellular, IEEE 802.16

Exam Tip

If the question says "key characteristics", structure each network as: **range · ownership · speed · technology · example**. Five short points × four networks = a full 8-mark answer.

30-Second Recap

PAN ≈ 1 room, LAN ≈ campus, MAN ≈ city, WAN ≈ country or world. Speed decreases and delay increases as you go from LAN to WAN. LAN is private; WAN uses public carriers.

1.10 Switching Overview, the Internet and Access Methods SUPPORTING TOPIC: Learn for short answers]

Exam Insight

Slides 53–57. Not asked yet, but Unit 4 expands switching. The WAN-vs-Internet distinction is a favourite short question.

- A **switch** connects at least two links and forwards data from one network segment to another. The two main types are **circuit-switched** (switch only forwards; no storage) and **packet-switched** (router stores and forwards each independent packet). Full detail is in Unit 4.
- The **Internet** is a **network of networks**: LANs and WANs connected through suitable hardware and software to work seamlessly.
- **WAN vs Internet**: a WAN is owned by a single organisation; the Internet is not owned by any one organisation.

Ways to access the Internet

Method	Details / example
Telephone networks	Dial-up services and DSL service over existing phone lines.
Cable network	Broadband delivered over the cable-TV network.
Wireless networks	Wi-Fi hotspots and mobile data.
Direct connection	A dedicated link from the organisation straight to an ISP.

UNIT 2 — Data Link Layer: Framing, Flow Control and Error Control

What this unit is about: how two directly connected devices cut the bit stream into frames (framing), avoid overwhelming each other (flow control) and recover from damaged or lost frames (error control with ARQ). Chapter 11, slides 59–116. This is your **highest-scoring unit**: ARQ appeared in 3 of 4 past papers and framing in 2 of 4.

2.1 Role of the Data Link Layer [★ ESSENTIAL FOR EXAM]

Exam Insight

Asked as: Dec 2025 Q1a — "Explain the key functions of the Application layer and Data Link layer" (8 marks). The data-link half is answered by the functions below.

Simple Analogy

Imagine two neighbouring post offices connected by one road. The **data link layer** is the van driver on that single road. His job: pack letters into labelled bags (**framing**), not deliver faster than the next office can sort (**flow control**), and re-send any bag that arrives torn or never arrives (**error control**). He is only responsible for **one hop**, not the entire journey.

Definition (write this in the exam)

The data link layer takes bits from the physical layer and packs them into **frames**, then delivers those frames reliably from **one node to the next node** on the same link. Flow control and error control together are called **data link control**.

Function	What it does	Example
Framing	Packs bits into frames so each frame is distinguishable from the next.	Putting a letter in an envelope.
Flow control	Restricts how much the sender may send before waiting for an acknowledgment.	A fast server must not flood a slow phone.
Error control	Detects damaged or lost frames and gets them retransmitted (ARQ).	A video-call app re-requesting a corrupted packet.
Physical addressing *	Adds sender and receiver link addresses to the frame header.	A MAC address in an Ethernet frame.
Access control *	Decides which device may use a shared link at a time.	Two laptops sharing one Wi-Fi channel.

* Starred functions are standard textbook functions of this layer; the slides focus on framing, flow control and error control.

2.2 Framing: Byte Stuffing and Bit Stuffing [💧 HIGH-YIELD: Frequently Asked]

Exam Insight

Asked as: Jan 2023 Q3B — "Illustrate with example the concept of bit stuffing" (6 marks). Dec 2025 Q2b — "Why is framing required at the data link layer? Give reasons and explain any three header fields of a data link frame" (6 marks). Question Bank Q30 (justify framing + differentiate bit/byte stuffing), Q31 (bit stuffing), Q32 (byte stuffing).

Frequency: 2 of 4 papers at 6 marks each. Learn the **why**, one **worked example** of each stuffing method, and the **comparison table**.

A. Why do we need framing?

Simple Analogy

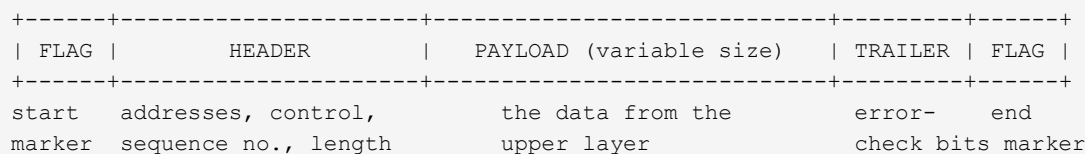
A continuous stream of bits is like a **long sentence with no spaces or full stops**:

"ITISRAININGTODAYTAKEUMBRELLA". Framing adds the full stops (flags) so the receiver knows where one message ends and the next begins. The slides use the postal system: the **envelope is the delimiter** that separates one letter from another.

#	Reason	Explained with an example
1	Marks frame boundaries	Without a start and end marker, the receiver cannot tell where frame 1 stops and frame 2 starts.
2	Synchronisation	The receiver locks onto the flag and knows exactly when a new frame begins.
3	Makes addressing possible	Each frame has a header with source and destination addresses, like the address on an envelope.
4	Per-frame error detection	The trailer carries check bits, so only the damaged frame is rejected, not the entire stream.
5	Per-frame flow and error control	ACK, NAK and sequence numbers refer to individual frames; ARQ would be impossible without frames.
6	Limits damage	One corrupted frame can be re-sent without re-sending everything.

B. Frame structure and header fields

A typical variable-size frame



Header / trailer field	Purpose	Example
Destination address	Tells which node should receive the frame.	Receiver's link address.
Source address	Tells who sent the frame, so a reply can be addressed.	Sender's link address.
Control / sequence number	Carries frame number, ACK number and type (data, ACK, NAK) used by flow and error control.	Frame number 0 or 1 in Stop-and-Wait ARQ.
Length / type	Tells how long the payload is, or which upper-layer protocol it belongs to.	"Payload = 46 bytes".
Trailer (error-check bits)	Lets the receiver detect corruption.	CRC or checksum value.

Model answer skeleton: "Why is framing required? Explain any three header fields" (6 marks)

- **Para 1 (3 marks):** Give 3 reasons: boundaries, synchronisation, per-frame addressing/error control. Add the envelope analogy.
- **Para 2 (3 marks):** Explain three header fields (destination address, source address, control/sequence number), one line each, with the frame diagram above.

C. Fixed-size vs variable-size framing

Type	How it works	Example
Fixed-size	Every frame has the same length, so no delimiter is needed; the size itself marks the boundary.	ATM cells (always 53 bytes).
Variable-size	Frames differ in length, so we must define where a frame ends. Two approaches: character-oriented (byte stuffing) and bit-oriented (bit stuffing).	Ethernet, PPP, HDLC.

D. Byte stuffing (character-oriented framing)

Definition (write this in the exam)

Byte stuffing is the process of adding **one extra byte** whenever there is a **flag** or **escape** character in the data. The extra byte is an **ESC** character inserted before it.

Simple Analogy

Writing a **quotation inside quotation marks**. If the quote itself contains a quotation mark, you put a backslash before it so the reader does not think the quote has ended. ESC is that backslash.

Step-by-Step Logic

1. Choose a special **FLAG** byte (frame delimiter) and an **ESC** byte.
2. **Sender:** scan the payload. Whenever you see FLAG or ESC inside the data, **insert an ESC before it**.
3. Add FLAG at the start and end of the frame.
4. **Receiver:** when it sees ESC, **remove the ESC** and treat the next byte as plain data. A FLAG **not** preceded by ESC is a real frame boundary.

Byte stuffing and unstuffing

```
Original payload :  A  FLAG  B  ESC  C

After STUFFING   :  A  ESC FLAG  B  ESC ESC  C
                   ^^^          ^^^
                   ESC added    ESC added before the ESC in data

Frame sent       :  FLAG | Header | A ESC FLAG B ESC ESC C | Trailer | FLAG

Receiver UNSTUFFS:  removes each protecting ESC -> A FLAG B ESC C
```

Real-World Example

PPP (Point-to-Point Protocol, used for dial-up and many modem links) uses byte stuffing with flag = 01111110 (0x7E) and escape = 0x7D.

Code snippet: byte stuffing / unstuffing

```
# Python: byte stuffing
FLAG, ESC = "F", "E"          # symbolic flag and escape characters

def byte_stuff(data):
    out = ""
    for ch in data:
        if ch in (FLAG, ESC):
            out += ESC          # insert escape before special byte
        out += ch
    return out

def byte_unstuff(data):
    out, i = "", 0
    while i < len(data):
        if data[i] == ESC:     # skip the escape, keep next byte as data
            i += 1
        out += data[i]
        i += 1
    return out

s = byte_stuff("AFBEC")       # A FLAG B ESC C
print(s)                      # AEFBEEC  (A ESC F B ESC E C)
print(byte_unstuff(s))        # AFBEC
```

E. Bit stuffing (bit-oriented framing)**Definition (write this in the exam)**

Bit stuffing is the process of adding **one extra 0** whenever **five consecutive 1s** appear in the data, so that the receiver does not mistake the data for the **flag pattern 0111110**.

Simple Analogy

The flag is a **secret knock: 0111110**. If your normal conversation accidentally contains five 1s in a row, you insert a "0" after them so nobody mistakes it for the secret knock. The receiver, knowing the rule, simply deletes the 0 that follows any five 1s.

Step-by-Step Logic

1. Flag = **0111110** (six consecutive 1s surrounded by 0s).
2. **Sender:** scan the data from left to right, counting consecutive 1s. After **five** consecutive 1s, **insert a 0**, then reset the count.
3. Put the flag at the start and the end of the frame.
4. **Receiver:** after **five** consecutive 1s, look at the next bit. If it is **0**, delete it (it was stuffed). If it is 1, it is part of a flag (frame boundary).

Worked Example (bit stuffing)

Data to send: **011011111101111100**

Bit stuffing and unstuffing

Data : 0 1 1 0 1 1 1 1 1 0 1 1 1 1 1 0 0
 ^{^^^^^^^^} ^{^^^^^^^^}
 1 1 1 1 1 (five 1s) 1 1 1 1 1 (five 1s)

Step 1: after the first five 1s -> insert 0

Step 2: after the next five 1s -> insert 0

Stuffed data : 0 1 1 0 1 1 1 1 1 [0] 1 0 1 1 1 1 1 [0] 1 0 0
 ^{^^^} ^{^^^} (inserted)

= 011011111010111110100 (19 bits became 21 bits)

Frame on the wire: 01111110 | 011011111010111110100 | 01111110
 FLAG stuffed data FLAG

Receiver: sees five 1s then a 0 -> deletes that 0 (twice) -> original data

Code snippet: bit stuffing / unstuffing

```
# Python: bit stuffing
def bit_stuff(bits):
    out, ones = "", 0
    for b in bits:
        out += b
        ones = ones + 1 if b == "1" else 0
        if ones == 5:
            # five consecutive 1s -> stuff a 0
            out += "0"
            ones = 0
    return out

def bit_unstuff(bits):
    out, ones, i = "", 0, 0
    while i < len(bits):
        out += bits[i]
        ones = ones + 1 if bits[i] == "1" else 0
        if ones == 5:
            # skip the stuffed 0 that follows
            i += 1
            ones = 0
        i += 1
    return out

d = "0110111110101111100"
s = bit_stuff(d)
print(s) # 011011111010111110100
print(bit_unstuff(s) == d) # True
```

Real-World Example

HDLC (High-level Data Link Control) uses bit stuffing with the 01111110 flag; it is the ancestor of many WAN link protocols.

F. Byte stuffing vs bit stuffing (QB Q30)

Feature	Byte stuffing	Bit stuffing
Framing style	Character-oriented (data is a sequence of characters).	Bit-oriented (data is a sequence of bits).
Unit of stuffing	One whole byte (ESC).	One bit (a 0).
Trigger	A FLAG or ESC byte appears in the data.	Five consecutive 1s appear in the data.

Feature	Byte stuffing	Bit stuffing
Flag	A special character such as 0x7E.	The bit pattern 01111110.
Overhead	Up to 1 extra byte (8 bits) per special byte.	1 extra bit per five 1s; usually lower.
Depends on character set?	Yes: assumes 8-bit characters.	No: works with any data and any code.
Example protocol	PPP	HDLC

⚠ Common Mistake

In bit stuffing, inserting the 0 after **six** 1s instead of **five**. The rule is: after exactly **five** consecutive 1s, always insert 0.

Forgetting to **reset the count** after inserting the 0.

Forgetting that in byte stuffing the ESC itself must also be escaped.

☑ 30-Second Recap

Framing = envelope: it marks where each frame starts and ends. A frame = FLAG + header + payload + trailer + FLAG.

Byte stuffing: put ESC before any FLAG or ESC in the data. Bit stuffing: put 0 after any five 1s.

The receiver reverses the process (unstuffing) to recover the original data.

2.3 Flow Control and Error Control [★ ESSENTIAL FOR EXAM]

📖 Exam Insight

Asked as: forms the theory half of every ARQ question and the data-link half of Dec 2025 Q1a. Slides 67–69.

💡 Simple Analogy

Flow control is like the **tap on a bucket**: you must not pour water faster than the bucket can take it, or it overflows. **Error control** is like **registered post**: if the parcel is damaged or lost, the sender is told and sends it again.

📖 Definition (write this in the exam)

Flow control is a set of procedures used to restrict the amount of data that the sender can send before waiting for an acknowledgment. **Error control** in the data link layer is based on **Automatic Repeat reQuest (ARQ)**, which means retransmission of data that was damaged or lost.

Feature	Flow control	Error control
Problem solved	Receiver is slower than sender; buffer overflow.	Frames are damaged, lost or duplicated.
Method	Sender stops and waits for ACK, or limits itself to a window of frames.	Detect error, then retransmit (ARQ) using ACKs, sequence numbers and timers.
Example	Sender at 100 Mbps, receiver processes 1 Mbps: without control, data is lost.	Frame 1 is corrupted by noise; the sender re-sends it after timeout.

The ARQ toolkit (four ingredients used by every protocol below)

- **Acknowledgment (ACK):** "I received it correctly."
- **Sequence numbers:** label each frame so duplicates and missing frames can be recognised.
- **Timer:** if no ACK arrives in time, assume loss and resend.
- **Copy of the frame:** the sender keeps it until the ACK arrives.

2.4 Noiseless Channel Protocols: Simplest and Stop-and-Wait [★ ESSENTIAL FOR EXAM]

 Exam Insight

Slides 72–82. These ideal-channel protocols are the stepping stones to Stop-and-Wait ARQ; the Question Bank (Q33, Q36) builds on them.

A. Simplest protocol (no flow control, no error control)

Assumes the receiver can process every frame immediately. The sender simply sends frames one after another. It is **unrealistic** but shows the basic design: the data link layer of the sender takes data from the network layer, makes a frame and sends it; the receiver extracts the data and hands it up.

Pseudocode: Simplest protocol

SENDER	RECEIVER
repeat forever:	repeat forever:
wait for data from network layer	wait for a frame to arrive
make frame; send it	extract data from frame
	deliver data to network layer

Real-World Example

Like a person dropping letters into a postbox and walking away, assuming the postman will always be ready.

B. Stop-and-Wait protocol (flow control only)

The sender sends **one frame**, then **stops and waits** for an acknowledgment before sending the next. This stops the sender from flooding a slower receiver. The channel is still assumed **noiseless**, so no frame is ever damaged.

Stop-and-Wait protocol (Example 11.2)

Sender	Receiver
--- Frame 1 ----->	Sender: waits
<----- ACK -----	Receiver: "ready for more"
--- Frame 2 ----->	
<----- ACK -----	

Sending 2 frames = 4 events at the sender (2 sends + 2 ACK arrivals)
+ 2 events at the receiver (2 frame arrivals)

Pseudocode: Stop-and-Wait protocol

SENDER canSend = true repeat forever: if canSend and data available: make frame; send it canSend = false if ACK arrives: canSend = true	RECEIVER repeat forever: wait for a frame deliver data to network layer send ACK
---	---

Simple Analogy

A **walkie-talkie conversation**: you say your sentence and say "over"; you do not speak again until the other person replies.

2.5 Stop-and-Wait ARQ (for Noisy Channels) [🔥 HIGH-YIELD: Frequently Asked]

Exam Insight

Asked as: Question Bank Q33 ("Explain in detail: Stop and Wait ARQ"), Q36 (compare with Go-Back-N). It is also the baseline that every GBN/SR answer is compared with. Slides 87–105.

Simple Analogy

You send a **registered letter** and wait for the signed receipt before posting the next one. If the receipt does not come back within a week, you assume the letter was lost and post a **photocopy** you kept. Numbering your letters (0, 1, 0, 1 ...) lets the receiver spot if the same letter arrives twice.

Definition (write this in the exam)

Stop-and-Wait ARQ is the simplest flow and error control mechanism. The sender sends a frame, **stops and waits for an acknowledgment**. If the ACK does not arrive before the timer expires, the sender **retransmits the frame**.

Key features

- The sender **keeps a copy** of the sent frame until the ACK arrives.
- The sender starts a **timer** when it sends a frame; if no ACK arrives in time, it **resends**.
- Frames and ACKs are numbered **alternately 0 and 1** (modulo-2 arithmetic). This identifies duplicates.
- The **ACK number is the number of the next frame expected**: receiving frame 0 correctly produces ACK 1.
- A damaged or lost frame is treated the same way by the receiver; if the frame is damaged or out of order the receiver **discards it silently** (no NAK).
- Control variables: **S** at the sender (number of the most recently sent frame, 0 or 1) and **R** at the receiver (number of the next frame expected, 0 or 1).

Four cases of operation

Case 1: Normal operation

Normal operation

```

Sender (S=0)                                Receiver (R=0)
|--- Frame 0 ----->| correct, deliver; R becomes 1
|<----- ACK 1 ----| "next expected is 1"
S becomes 1
|--- Frame 1 ----->| correct, deliver; R becomes 0
|<----- ACK 0 ----|

```

Case 2: Lost or damaged frame

Lost frame: recovered by timeout

```

Sender                                Receiver
|--- Frame 0 -----X                (frame lost / corrupted)
| [timer running .....]              (receiver silent)
| TIMEOUT!
|--- Frame 0 (copy resent) ----->| accepted
|<----- ACK 1 ----|

```

Case 3: Lost ACK (why numbering matters)

Lost ACK: duplicate detected by sequence number

```

Sender                                Receiver
|--- Frame 0 ----->| accepted, R becomes 1
|      X<----- ACK 1 --| ACK lost on the way
| TIMEOUT! (sender cannot tell if frame or ACK was lost)
|--- Frame 0 (resent) ----->| DUPLICATE: R expects 1, got 0
|                                | discard data, re-send ACK 1
|<----- ACK 1 ----|
|--- Frame 1 ----->| continue normally

```

⚠ Without sequence numbers

In Case 3 the receiver would accept the resent Frame 0 as **new** data, so the same data would be delivered **twice**. That is exactly why frames are numbered.

Case 4: Delayed ACK and a lost next frame

Delayed ACK: numbered ACKs prevent a lost frame being missed

```

Sender                                Receiver
|--- Frame 0 ----->| accepted, R becomes 1
|      ..... ACK 1 is delayed .....|
| TIMEOUT!
|--- Frame 0 (resent) ----->| duplicate: discard, ACK 1 again
|<--- ACK 1 (the delayed one) arrives |
|--- Frame 1 -----X                (frame 1 is lost)
|<--- ACK 1 (second copy) arrives    |
| Sender is now waiting for ACK 0 (S=1), so it IGNORES this stale ACK 1
| TIMEOUT -> resend Frame 1 ----->| accepted

```

 Exam Tip

Memorise the slide's two key notes: **numbered frames prevent duplicate frames being kept**, and **numbered ACKs are needed when an ACK is delayed and the next frame is lost**.

Pseudocode

Pseudocode: Stop-and-Wait ARQ SENDER

```

S = 0; canSend = true
repeat forever:
  if canSend and data available:
    frame = (seq=S, data)
    keep copy; send(frame)
    start timer; canSend = false
  on ACK(n) arrival (not corrupted):
    if n == (S+1) mod 2:
      stop timer; discard copy
      S = (S+1) mod 2; canSend = true
    # else: stale ACK, ignore
  on timer expiry:
    send(copy); restart timer

```

Pseudocode: Stop-and-Wait ARQ RECEIVER

```

R = 0
on frame arrival:
  if frame is corrupted: discard (silent)
  else:
    if frame.seq == R:
      deliver data; R = (R+1) mod 2
    send ACK(R) # always ack "next expected", even for duplicates

```

Efficiency: why Stop-and-Wait wastes the link

Worked Example (Example 11.4)

Line bandwidth = 1 Mbps; one bit takes 20 ms to make a round trip; frames are 1000 bits.

Bandwidth-delay product = bandwidth × round-trip time = $1,000,000 \times 0.020 = 20,000$ bits. In the time one frame goes out and its ACK comes back, the link could have carried 20,000 bits.

The sender sends only 1000 bits, so **utilisation** = $1000 / 20,000 = 5\%$.

Conclusion: for a link with high bandwidth or long delay, Stop-and-Wait wastes capacity. The fix is to send many frames before waiting (pipelining), described next.

30-Second Recap

Send one frame, wait for ACK. Timer expiry means resend the stored copy.

Sequence numbers 0/1; ACK number = next expected frame; receiver is silent on damaged frames.

Four cases: normal, lost frame, lost ACK, delayed ACK. Weakness: low link utilisation.

2.6 Pipelining and the Sliding Window **ESSENTIAL FOR EXAM**

Exam Insight

Asked as: supports Q34–39 and the numericals in 2.10. Slides 105–109.

Simple Analogy

A **factory conveyor belt**. In Stop-and-Wait, a worker places one box, waits until the customer says "received", and only then places the next box. With **pipelining** many boxes ride the belt at the same time, so the belt is always busy. The **window** is the number of boxes allowed on the belt that have not yet been confirmed.

- **Pipelining**: a task is begun before the previous task has ended. Stop-and-Wait has **no** pipelining; sliding-window protocols **do**.
- **Outstanding frames**: frames that have been sent but not yet acknowledged.
- The sender may send up to **W frames** and must keep a copy of each outstanding frame until its ACK arrives. The set of allowed sequence numbers is the **sliding window**.
- Both **Go-Back-N** and **Selective Repeat** are sliding-window ARQ protocols.

The sender window (m = 3 bits, sequence numbers 0–7)

 The window slides forward as ACKs arrive

```
sequence numbers: 0 1 2 | 3 4 5 6 7 0 1 | 2 3 ...
                  acked | SENDER WINDOW | not yet allowed
                  (done) | (size 7 for GBN) |
                  ^SF      ^SL
SF = first outstanding frame    SL = last frame the window allows
```

An ACK arrives for frame 3 and 4 (cumulative ACK 5) -> window SLIDES forward:

```
sequence numbers: 0 1 2 3 4 | 5 6 7 0 1 2 3 | 4 ...
                  ^SF      ^SL
```

Variable	Meaning
S	Sequence number of the most recently sent frame.
SF	Sequence number of the first (oldest) outstanding frame, i.e. the left edge of the window.
SL	Sequence number of the last frame in the window, i.e. the right edge.
R	Sequence number of the frame the receiver expects next.

Worked Example (Example 11.5): benefit of pipelining

Same link as before: bandwidth-delay product = 20,000 bits, frames of 1000 bits.

A protocol that can send up to **15 frames** before waiting sends $15 \times 1000 = 15,000$ bits per round trip.

Utilisation = $15,000 / 20,000 = 75\%$ (up from 5%). With damaged frames the utilisation would be lower because frames must be resent.

2.7 Go-Back-N ARQ [🔥 HIGH-YIELD: Frequently Asked]

Exam Insight

Asked as: Jan 2023 Q1B and Jan 2025 Q2c use Go-Back-N as the comparison baseline. Question Bank Q34 ("Explain in detail: Go-Back-N ARQ"), Q36, Q37, Q39. Slides 109–116.

Simple Analogy

A **teacher dictating numbered sentences**. A student misses sentence 3. The teacher says "stop, go back to sentence 3" and re-reads sentences 3, 4, 5, 6 **even though the student had already written 4, 5 and 6 correctly**. The student crossed out 4, 5, 6 earlier because they were out of order. That wasted effort is the weakness of Go-Back-N.

Definition (write this in the exam)

In **Go-Back-N ARQ** the sender may send up to $N = 2^m - 1$ frames without waiting for ACKs. If a frame is lost or damaged, the receiver **discards it and every frame that follows**; when the timer of the oldest unacknowledged frame expires, the sender **goes back and resends that frame and all frames after it**.

Key features

- **Sender window size = $2^m - 1$** (for $m = 3$: sequence numbers 0–7, window size 7).
- **Receiver window size = 1**. The receiver accepts only the frame it expects (R) and slides one slot at a time.
- **One timer** is used, for the **first outstanding frame**.
- The receiver sends a positive ACK if a frame arrives safe **and in order**. If a frame is damaged or out of order, the receiver is **silent** and **discards** it and all following frames.
- **Cumulative ACK**: the receiver need not acknowledge every frame; one ACK can confirm several (ACK 2 makes ACK 1 unnecessary).
- Correctly received out-of-order frames are **not buffered**. This is the main disadvantage.

Scenario A: normal operation with cumulative ACK

Go-Back-N: cumulative acknowledgment

Sender	Receiver
--- Frame 0 ----->	accept (R=1)
--- Frame 1 ----->	accept (R=2)
--- Frame 2 ----->	accept (R=3)
<----- ACK 3 -----	ONE cumulative ACK covers 0, 1, 2
window slides: SF moves from 0 to 3	

Scenario B: a frame is lost (the classic exam diagram)

Go-Back-N: lost frame (1 lost, 3 retransmitted)

Sender	Receiver
--- Frame 0 ----->	accept (R=1)
--- Frame 1 ----->	accept (R=2)
<----- ACK 2 -----	cumulative ACK for 0 and 1
--- Frame 2 -----X	LOST
--- Frame 3 ----->	expected 2, got 3 -> DISCARD, silent
--- Frame 4 ----->	expected 2, got 4 -> DISCARD, silent
... timer of Frame 2 expires ...	
=== GO BACK: resend Frame 2 and everything after it ===	
--- Frame 2 (again) ----->	accept (R=3)
--- Frame 3 (again) ----->	accept (R=4)
--- Frame 4 (again) ----->	accept (R=5)
<----- ACK 5 -----	

🔗 Worked Example (slide 114)

The sender has sent frames up to **6**. The timer for **frame 3** expires because frame 3 was never acknowledged.

The sender goes back and resends **frames 3, 4, 5 and 6** (4 frames) even though only frame 3 was actually lost. Frames 4, 5, 6 may have arrived fine but were discarded by the receiver.

Why is the sender window limited to $2^m - 1$?**🔗 Worked Example: $m = 3$, sequence numbers 0–7**

If the window were 8: the sender sends frames 0–7 and the receiver accepts all and expects **0** again. Suppose all ACKs are lost. After the timeout the sender resends **frame 0**. The receiver expects 0, so it **accepts the old frame as new data**. Error!

With window 7: the sender sends 0–6; the receiver now expects **7**. The resent frame 0 is not 7, so it is correctly **discarded** as a duplicate. ✓

📋 Pseudocode: Go-Back-N SENDER

```
Sw = 2^m - 1; Sf = 0; Sn = 0      # first outstanding, next to send
on data from network layer:
    if (Sn - Sf) mod 2^m < Sw:
        send frame(seq=Sn); keep copy
        if no timer running: start timer
        Sn = (Sn + 1) mod 2^m
on ACK(ackNo) arrival:
    while Sf != ackNo:
        purge copy of Sf; Sf = (Sf + 1) mod 2^m
    if frames still outstanding: restart timer else stop
on timer expiry:
    restart timer
    resend ALL frames from Sf up to Sn-1      # "go back N"
```

📋 Pseudocode: Go-Back-N RECEIVER

```
Rn = 0
on frame arrival:
    if corrupted or seq != Rn:
        discard (silent)
    else:
        deliver data
        Rn = (Rn + 1) mod 2^m
        send ACK(Rn)
```

Advantages	Disadvantages
- Efficient: pipelining keeps the link busy. - Simple receiver (window = 1, no buffer for out-of-order frames). - Cumulative ACKs reduce ACK traffic. - Much better utilisation than Stop-and-Wait.	- One error causes many frames to be resent (wasteful). - Good frames received after an error are discarded. - Poor on noisy or long-delay links. - Window limited to $2^m - 1$.

☑ 30-Second Recap

GBN = sender window $2^m - 1$, receiver window 1, one timer, cumulative ACK.

On loss: receiver discards all later frames; sender resends the lost frame **and every frame after it**.

Weakness: it re-sends frames that were already received correctly.

2.8 Selective Repeat ARQ [🔥 HIGH-YIELD: Frequently Asked]

Exam Insight

Asked as: Jan 2022 Q4-OR ("How does Selective-Repeat ARQ provide error control and flow control? Illustrate with an example", 8 marks). Also the "better" protocol in Jan 2023 Q1B and Jan 2025 Q2c. Question Bank Q35, Q38.

Important: your slides only **mention** Selective Repeat (slides 83, 105, 107). This section follows the standard textbook treatment, so read it carefully.

Simple Analogy

Same teacher, but now each student has a **notebook with a numbered page per sentence**. If sentence 3 is missed, the student keeps writing 4 and 5 in their pages and says "I am missing sentence 3". The teacher re-reads **only sentence 3**. The student then reads out 3, 4, 5 in order. No work is wasted.

Definition (write this in the exam)

In **Selective Repeat ARQ** the receiver **accepts and buffers** out-of-order frames that fall inside its window, and only the **damaged or lost frame is retransmitted**. Frames are delivered to the network layer **in order**.

Key features

- **Sender window size = $2^{(m-1)}$** and **receiver window size = $2^{(m-1)}$** (for $m = 3$: both 4).
- The receiver **buffers** correctly received frames that arrive out of order, within its window.
- The receiver acknowledges frames **individually** (ACK n confirms frame n) and sends a **NAK** to request a specific missing frame.
- The sender keeps a **timer for each outstanding frame** and resends only the one that timed out or was NAKed.
- When the missing frame finally arrives, the receiver delivers it together with the buffered frames, **in order**, and slides its window.

Scenario: frame 2 is lost

Selective Repeat: lost frame (1 lost, 1 retransmitted)

Sender	Receiver
--- Frame 0 ----->	accept, deliver, ACK 0
--- Frame 1 ----->	accept, deliver, ACK 1
--- Frame 2 -----X	LOST
--- Frame 3 ----->	out of order but INSIDE window:
	BUFFER it, send NAK 2, ACK 3
--- Frame 4 ----->	BUFFER it, send ACK 4
<----- NAK 2 -----	
--- Frame 2 (resend ONLY this) ----->	accept; now deliver 2, 3, 4 in order
<----- ACK 2 -----	

Frames resent: 1 (Go-Back-N would have resent 3)

Why is the window limited to $2^{(m-1)}$?

Worked Example: $m = 2$, sequence numbers 0–3

If the window were 3: the sender sends 0, 1, 2. The receiver accepts all three and its window moves to expect 3, 0, 1. All ACKs are lost. The sender resends frame 0. The receiver's new window contains 0, so it **accepts the old frame as a brand-new frame**. Error!

With window 2: the sender sends 0, 1. The receiver's window moves to 2, 3. The resent frame 0 is **outside** the window, so it is recognised as a duplicate and just re-acknowledged. ✓

Pseudocode: Selective Repeat SENDER

```

Sw = 2^(m-1); Sf = 0; Sn = 0
on data from network layer:
    if (Sn - Sf) mod 2^m < Sw:
        send frame(Sn); keep copy
        start a timer for this frame
        Sn = (Sn + 1) mod 2^m
on ACK(k):
    mark frame k acknowledged; stop its timer
    while frame Sf is acknowledged:
        slide window: Sf = (Sf + 1) mod 2^m
on NAK(k):      resend frame k only
on timeout(k):  resend frame k only; restart its timer

```

Pseudocode: Selective Repeat RECEIVER

```

Rw = 2^(m-1); Rn = 0; buffer = {}
on frame(seq) arrival:
    if corrupted: discard (may send NAK(Rn))
    else if seq inside [Rn, Rn+Rw-1]:
        if seq != Rn: send NAK(Rn) (once)
        store frame in buffer[seq]
        send ACK(seq)
        while buffer[Rn] exists:
            deliver(buffer[Rn]); Rn = (Rn + 1) mod 2^m
    else:
        send ACK(seq)      # old duplicate: re-acknowledge

```

Advantages	Disadvantages
- Only the lost frame is retransmitted: saves bandwidth. - No good frame is thrown away. - Best on noisy or long-delay links. - Uses NAK for fast recovery.	- More complex sender and receiver. - Receiver needs a buffer and a sorting mechanism. - A timer (or tracking state) for each frame. - Window limited to $2^{(m-1)}$ (half of GBN).

Real-World Example

Real TCP implementations use selective acknowledgments (SACK) following the same idea: retransmit only what is missing.

30-Second Recap

SR = sender window $2^{(m-1)}$, receiver window $2^{(m-1)}$, buffers out-of-order frames, individual ACK/NAK, one timer per frame.

Only the missing frame is resent; frames are delivered in order.

Cost: more memory, more logic.

2.9 Selective Repeat vs Go-Back-N: Full Comparison [🔥 HIGH-YIELD: Frequently Asked]

Exam Insight

🔥 **TOP PRIORITY — the most repeated question in your papers.** Jan 2023 Q1B (8 marks, BL5) and Jan 2025 Q2c (6 marks): "Why does Selective-Repeat ARQ give better performance than Go-Back-N ARQ? Justify your answer with a proper diagram and critical comments." Jan 2022 Q4 asked a variant. Question Bank: Q36 (Stop-and-Wait vs GBN), Q37 (GBN vs SR).

Model answer skeleton (6–8 marks)

- Intro (1 line):** both are sliding-window ARQ protocols that pipeline frames; they differ in how they recover from an error.
- Diagram:** draw the same scenario (frame 2 lost) for both protocols side by side (see below).
- Justification:** explain that GBN discards good frames and resends everything after the error, while SR buffers them and resends only the lost frame.
- Table:** 6–8 rows of differences.
- Critical comments:** SR is better on noisy / long-delay links but costs more memory and complexity; GBN is better when errors are rare and receivers are simple. Give your verdict.

Same error, two behaviours

Retransmission comparison

GO-BACK-N (frame 2 lost)

S: F0 F1 F2(X) F3 F4
 R: ok ok -- drop drop
 timeout ->
 S: F2 F3 F4 (3 frames resent)
 R: ok ok ok

Frames re-sent: 3

SELECTIVE REPEAT (frame 2 lost)

S: F0 F1 F2(X) F3 F4
 R: ok ok -- buffer buffer + NAK 2
 NAK/timeout ->
 S: F2 (1 frame resent)
 R: delivers 2,3,4 in order

Frames re-sent: 1

Feature	Go-Back-N ARQ	Selective Repeat ARQ
Frames resent after an error	The damaged frame and all frames after it.	Only the damaged / lost frame.
Sender window size	$2^m - 1$	$2^{(m-1)}$
Receiver window size	1	$2^{(m-1)}$ (same as sender)
Out-of-order frames	Discarded (not buffered).	Buffered until the gap is filled.
Acknowledgment	Cumulative ACK (one ACK confirms many).	Individual ACK per frame, plus NAK for missing frame.
Timers	One timer (for the first outstanding frame).	One timer per outstanding frame.
Receiver complexity	Simple: no sorting, no buffer.	Complex: buffer plus reordering.
Bandwidth efficiency	Lower on noisy links (resends good frames).	Higher (no wasted retransmissions).
Best suited for	Low error rate, simple devices.	High error rate or long delay (e.g. satellite).

Three-way comparison with Stop-and-Wait ARQ (QB Q36, Q37)

Feature	Stop-and-Wait ARQ	Go-Back-N ARQ	Selective Repeat ARQ
Sender window	1	$2^m - 1$	2^{m-1}
Receiver window	1	1	2^{m-1}
Sequence numbers	0, 1 (1 bit)	0 to $2^m - 1$	0 to $2^m - 1$
Pipelining	No	Yes	Yes
Link utilisation	Very low	Good	Best
Retransmission	The one frame	Lost frame + all following	Only the lost frame
Complexity	Lowest	Medium	Highest

Code snippet: retransmission count

```
# Python: how many frames are re-sent after ONE loss?
def resent_after_loss(last_sent, lost):
    gbn = last_sent - lost + 1      # lost frame + everything sent after it
    sr = 1                        # only the lost frame
    return gbn, sr

print(resent_after_loss(last_sent=6, lost=3))  # (4, 1) -> matches slide 114
print(resent_after_loss(last_sent=4, lost=2))  # (3, 1) -> diagram above
```

Critical comments (the BL5 marks)

SR wins on **performance** because it never wastes bandwidth re-sending good frames, which matters most on **noisy or high-delay** links.

But SR **costs more**: receiver buffer memory, sorting logic and per-frame timers. If errors are rare, GBN performs almost the same with far simpler hardware.

Both beat Stop-and-Wait because pipelining keeps the link busy. Your **verdict**: choose SR when channel quality is poor or delay is large; otherwise GBN is adequate.

30-Second Recap

Same pipelining; different error recovery. GBN = resend the lost frame and everything after. SR = resend only the lost frame.

GBN: window $2^m - 1$ / 1, cumulative ACK, discards. SR: window 2^{m-1} / 2^{m-1} , buffers, NAK.

2.10 Numericals: Window Sizes and Link Utilisation [ESSENTIAL FOR EXAM]

Exam Insight

Asked as: Question Bank Q38 (Selective Repeat, maximum sequence number 15) and Q39 (Go-Back-N, maximum sequence number 15). Easy 1–3 marks if you know the three formulas.

Protocol	Sender window	Receiver window
Stop-and-Wait ARQ	1	1
Go-Back-N ARQ	$2^m - 1$	1
Selective Repeat ARQ	2^{m-1}	2^{m-1}

Here **m** is the number of bits used for the sequence number, so there are 2^m sequence numbers (0 to $2^m - 1$). **Maximum sequence number 15 means 0–15, i.e. 16 numbers, so $m = 4$.**

Step-by-Step Logic

1. Find the number of sequence numbers: max sequence number + 1.
2. Find m: $2^m = \text{that number}$.
3. Apply the formula for the protocol.

Worked Examples

QB Q38 (SR, max seq 15): 16 numbers $\rightarrow m = 4 \rightarrow \text{sender window} = \text{receiver window} = 2^{4-1} = 8$.

QB Q39 (GBN, max seq 15): $m = 4 \rightarrow \text{sender window} = 2^4 - 1 = 15$; receiver window = 1.

GBN with 3-bit sequence numbers (slide 109): sender window = $2^3 - 1 = 7$.

SR with max seq 7: 8 numbers $\rightarrow m = 3 \rightarrow \text{both windows} = 2^2 = 4$.

5-bit sequence numbers: GBN sender = 31; SR windows = 16.

Link utilisation

Definition (write this in the exam)

Bandwidth-delay product (BDP) = bandwidth $\times$ round-trip time (in bits).

Utilisation = (bits sent per round trip) / BDP, capped at 100%.

Stop-and-Wait: bits per round trip = one frame. Sliding window: bits per round trip = window size $\times$ frame size.

Practice problem with solution

Link = 2 Mbps, round-trip time = 40 ms, frame = 2000 bits.

BDP = $2,000,000 \times 0.040 = 80,000$ bits.

Stop-and-Wait utilisation = $2000 / 80,000 = 2.5\%$.

Go-Back-N with window 15: $15 \times 2000 / 80,000 = 37.5\%$.

Lesson: pipelining multiplies utilisation by the window size (until 100%).

Code snippet: formulas as code

```
# Python: window sizes and utilisation
def windows(max_seq):
    m = (max_seq + 1).bit_length() - 1      # 2**m = max_seq + 1
    return {"GBN": (2**m - 1, 1), "SR": (2**(m-1), 2**(m-1))}

def utilisation(bw_bps, rtt_s, frame_bits, window=1):
    bdp = bw_bps * rtt_s
    return min(1.0, window * frame_bits / bdp)

print(windows(15))                        # {'GBN': (15, 1), 'SR': (8, 8)}
print(utilisation(1e6, 0.020, 1000))      # 0.05  -> 5%
print(utilisation(1e6, 0.020, 1000, 15))  # 0.75  -> 75%
```

Common Mistake

Using the **max sequence number as m**. If the max sequence number is 15, m is **4**, not 15.

Mixing the formulas: GBN = $2^m - 1$, SR = $2^{(m-1)}$. GBN receiver is always **1**.

UNIT 3 — Data and Signals

What this unit is about: how data becomes signals, how signals are described (amplitude, frequency, phase, bandwidth), how fast we can send bits (bit rate) and what damages signals on the way (impairment). Chapter 3, slides 117–196. The unit is **numerical-heavy**: practise every worked example.

3.1 Analog and Digital Data and Signals [■ SUPPORTING TOPIC: Learn for short answers]

Exam Insight

Asked as: Question Bank Q1(i–l): define analog data, analog signal, digital data, digital signal (1–3 marks each). Slides 119–123.

Simple Analogy

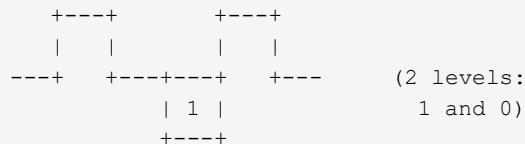
Analog = a ramp, digital = a staircase. On a ramp you can stand at any height (continuous). On a staircase you can only stand on a step (discrete). Think of an old clock with sweeping hands (analog) versus a digital watch showing 10:41 (digital).

Analog vs digital signal

ANALOG SIGNAL (continuous)



DIGITAL SIGNAL (discrete levels)



Term	Definition	Example
Analog data	Continuous information; takes continuous values.	Human voice, temperature.
Digital data	Information with discrete states; takes discrete values.	Data stored in a computer.
Analog signal	Can have an infinite number of values in a range.	Sound wave from a microphone.
Digital signal	Can have only a limited number of values.	Voltage 0 V / 5 V in a wire.

Exam Tip

In data communications we **commonly use periodic analog signals and non-periodic digital signals** (slide 123). Data must be transformed into electromagnetic signals before it can be transmitted.

3.2 The Sine Wave: Amplitude, Period, Frequency, Phase, Wavelength [■ SUPPORTING TOPIC: Learn for short answers]

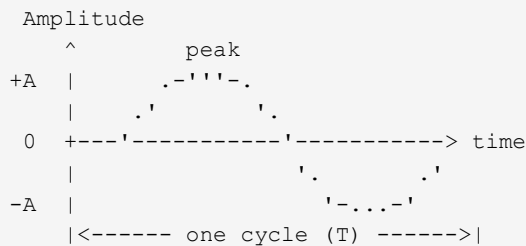
Exam Insight

Slides 124–143. Not asked directly, but you need these to understand bandwidth. Unit-conversion examples can appear as 2–3 mark numericals.

Simple Analogy

A sine wave is a **playground swing**. How high it goes = amplitude. How long one full swing takes = period. How many swings per second = frequency. Where in its swing it started = phase.

A sine wave



Period T = time for ONE full cycle

Frequency f = cycles per second = $1/T$

Phase = how far the wave is shifted from time 0

Parameter	Meaning	Unit	Formula
Peak amplitude	Absolute value of the signal's highest intensity (strength).	Volts	—
Period (T)	Time to complete one cycle.	Seconds	$T = 1/f$
Frequency (f)	Number of cycles per second; rate of change with respect to time.	Hertz (Hz)	$f = 1/T$
Phase	Position of the waveform relative to time 0.	Degrees or radians	1 cycle = $360^\circ = 2\pi$ rad
Wavelength (λ)	Distance a signal travels in one period.	Metres	$\lambda = v \times T = v / f$

- **Frequency and period are inverses.** High frequency = changes quickly; low frequency = changes slowly.
- If a signal does not change at all, its frequency is **zero**. If it changes instantaneously, its frequency is **infinite**.
- Units: 1 kHz = 10^3 Hz, 1 MHz = 10^6 Hz; 1 ms = 10^{-3} s, 1 μ s = 10^{-6} s.

Worked Examples (slides 133–140)

Ex 3.3: household power is 60 Hz. Period $T = 1/60 = 0.0166$ s = **16.6 ms**.

Ex 3.4: express 100 ms in microseconds. $100 \text{ ms} = 100 \times 10^{-3} \text{ s} = 100 \times 10^{-3} \times 10^6 \mu\text{s} = \mathbf{10^5 \mu\text{s}}$.

Ex 3.5: period = 100 ms. $f = 1 / 0.1 \text{ s} = 10 \text{ Hz} = \mathbf{0.01 \text{ kHz}}$.

Ex 3.6: a wave offset by 1/6 cycle. Phase = $360^\circ / 6 = \mathbf{60^\circ} = 60 \times (2\pi/360) = \mathbf{\pi/3 \approx 1.047 \text{ rad}}$.

Ex 3.1: home power is quoted as 110–120 V but its **peak** is higher: peak = $\sqrt{2} \times \text{rms} \approx 1.414 \times \text{rms}$ (rms 120 V gives about 170 V peak).

Code snippet: period, phase, wavelength

```
# Python: sine wave conversions
import math
f = 60                                # Hz
T = 1 / f                             # period in seconds
print(round(T * 1000, 1))             # 16.7 ms (16.6 when truncated)

phase_deg = 360 / 6                   # 1/6 cycle
print(phase_deg, math.radians(phase_deg)) # 60.0 1.0471...

v = 3e8                               # speed of light, m/s (assumed)
print(v / f)                          # wavelength = 5,000,000 m
```

3.3 Time Domain vs Frequency Domain [■ SUPPORTING TOPIC: Learn for short answers]

Exam Insight

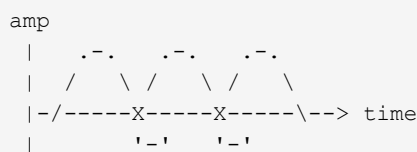
Slides 142–146. This is the "language" used to draw spectra in the bandwidth numericals (3.5), which are high-yield.

Simple Analogy

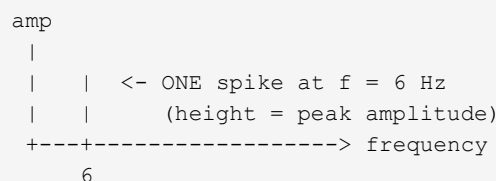
Music. The **time-domain** view is the sheet-music rolling in time (what note when). The **frequency-domain** view is a **graphic equalizer**: bars showing how much of each pitch is present. A single pure note is just one bar.

A sine wave in both domains

TIME DOMAIN (amplitude vs time)



FREQUENCY DOMAIN (amplitude vs frequency)



A complete sine wave in time = ONE single spike in frequency.

Real-World Example

Three sine waves with different frequencies and amplitudes appear as **three spikes** in the frequency domain, which is much more compact than drawing three tangled waves in the time domain (Example 3.7).

3.4 Composite Signals [■ SUPPORTING TOPIC: Learn for short answers]

Exam Insight

Slides 147–154. Foundation for bandwidth. Key line to memorise: periodic composite signal gives a discrete spectrum, non-periodic gives a continuous spectrum.

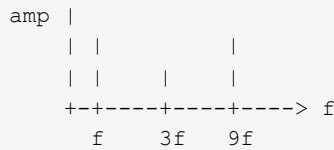
Simple Analogy

A **smoothie**: the final drink (composite signal) is a mix of fruits (simple sine waves). **Fourier analysis** is the machine that separates the smoothie back into its fruits, telling you how much of each fruit (amplitude) and what kind (frequency).

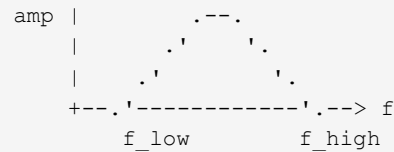
- A **single-frequency sine wave is not useful** for data communication; we send a **composite signal**, made of many sine waves.
- **Fourier analysis**: any composite signal is a combination of simple sine waves with different frequencies, amplitudes and phases.
- If the composite signal is **periodic**, decomposition gives **discrete frequencies** (separate spikes).
- If it is **non-periodic**, decomposition gives **continuous frequencies** (a smooth curve).

Spectrum of periodic vs non-periodic signals

PERIODIC composite
(discrete spikes)



NON-PERIODIC composite
(continuous spectrum)



Type	Example	Why
Periodic composite	Three alarm systems ringing at different frequencies (Ex 3.8).	The pattern repeats forever.
Non-periodic composite	A microphone signal while you speak a word or two (Ex 3.9).	You never repeat the word with exactly the same tone.

3.5 Bandwidth and Frequency-Spectrum Numericals **HIGH-YIELD: Frequently Asked**

Exam Insight

Asked as: Jan 2022 Q8 — "A non-periodic composite signal has a bandwidth of 100 kHz, middle frequency 150 kHz, peak amplitude 10 V; the two extreme frequencies have amplitude 0. Draw the frequency domain" (3 marks). Question Bank Q19 (five sine waves) and Q20 (non-periodic signal), 2–4 marks each.

Why high-yield: the Jan 2022 question is Q20 with different numbers. Learn the template once and you can solve any version.

Definition (write this in the exam)

The **bandwidth of a composite signal** is the **difference between the highest and the lowest frequencies** contained in that signal: $B = f_{\text{high}} - f_{\text{low}}$

Simple Analogy

Bandwidth is the **width of a road**. The more lanes (frequencies) the road has, the more traffic (information) it can carry at once. A 200 kHz FM station is a wider road than a 10 kHz AM station, which is why FM sounds richer.

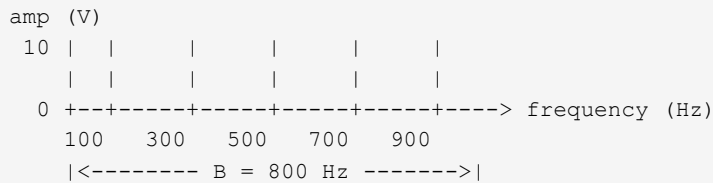
Template for every bandwidth numerical

1. Write the formula: $B = f_{\text{h}} - f_{\text{l}}$.
2. **Periodic signal** → list the given frequencies; draw **separate vertical spikes**.
3. **Non-periodic signal** → find f_{l} and f_{h} from the **middle frequency**: $f_{\text{l}} = \text{middle} - B/2$, $f_{\text{h}} = \text{middle} + B/2$. Draw a **smooth curve** with amplitude 0 at the ends and the peak at the middle.
4. Label both axes (frequency, amplitude), the peak value and the bandwidth arrow.

Ex 3.10 / QB Q19: five sine waves at 100, 300, 500, 700, 900 Hz

$B = f_{\text{h}} - f_{\text{l}} = 900 - 100 = 800 \text{ Hz}$. The spectrum has only five spikes, each 10 V high.

Spectrum for Example 3.10



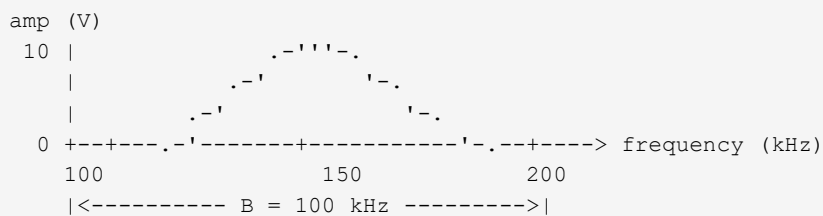
Ex 3.11: periodic signal, $B = 20$ Hz, highest frequency 60 Hz

$f_l = f_h - B = 60 - 20 = 40$ Hz. A periodic signal containing all integer frequencies of the same amplitude gives spikes at 40, 41, 42 ... 60 Hz.

Jan 2022 Q8: $B = 100$ kHz, middle frequency 150 kHz, peak 10 V

$f_l = 150 - 100/2 = 100$ kHz; $f_h = 150 + 50 = 200$ kHz. Amplitude is 0 at both extremes and **10 V at 150 kHz**.

Frequency domain for Jan 2022 Q8 (non-periodic: smooth curve)



Ex 3.12 / QB Q20: $B = 200$ kHz, middle 140 kHz, peak 20 V

$f_l = 140 - 100 = 40$ kHz; $f_h = 140 + 100 = 240$ kHz; peak **20 V at 140 kHz**, 0 at 40 and 240 kHz.

Real-world bandwidth examples (slides 163–165)

Signal	Bandwidth	Note
AM radio station (USA)	10 kHz per station	Total AM band: 530 to 1700 kHz.
FM radio station (USA)	200 kHz per station	Total FM band: 88 to 108 MHz.
Old analog black-and-white TV	5.5125 MHz	$525 \times 700 = 367,500$ pixels $\times$ 30 screens/s = 11,025,000 pixels/s; 2 pixels per cycle gives 5,512,500 Hz.

Practice problems (solve, then check)

P1. A periodic signal has components at 200, 400, 600, 800 Hz. $B = 800 - 200 = 600$ Hz.

P2. Non-periodic, $B = 50$ kHz, middle frequency 175 kHz. $f_l = 175 - 25 = 150$ kHz, $f_h = 200$ kHz.

P3. Periodic, $f_h = 5$ kHz, $B = 3.1$ kHz. $f_l = 5 - 3.1 = 1.9$ kHz.

Code snippet: bandwidth numericals

```
# Python: bandwidth helper
def bandwidth_periodic(freqs):
    return max(freqs) - min(freqs)

def nonperiodic_range(bandwidth, middle):
    return middle - bandwidth / 2, middle + bandwidth / 2

print(bandwidth_periodic([100, 300, 500, 700, 900])) # 800
print(nonperiodic_range(100, 150)) # (100.0, 200.0) Jan 2022 Q8
print(nonperiodic_range(200, 140)) # (40.0, 240.0) QB Q20
```

⚠ Common Mistake

Drawing **spikes** for a **non-periodic signal**. Non-periodic = continuous = a **curve**.

Forgetting that the extremes have **amplitude 0** when the question says so.

Writing bandwidth as the **highest** frequency instead of **highest – lowest**.

✓ 30-Second Recap

$B = f_h - f_l$. Periodic → spikes; non-periodic → curve. For a non-periodic signal: $f_l = \text{middle} - B/2$, $f_h = \text{middle} + B/2$.

3.6 Digital Signals: Bit Rate, Bit Length and Baseband [★ ESSENTIAL FOR EXAM]**📖 Exam Insight**

Asked as: Question Bank Q21 ("A digital signal has eight levels. How many bits per level?", 1–2 marks) and Q22 ("Download text at 100 pages per minute: required bit rate?", 1–3 marks). Slides 166–179.

🧠 Simple Analogy

A **traffic light** with 2 colours can send 1 yes/no bit per change. A light with **8 colours** can send 3 bits per change (because $2^3 = 8$). More levels, more bits per signal change.

📖 Definition (write this in the exam)

Bit rate is the number of bits sent in **one second**, measured in **bits per second (bps)**.

📄 Digital signals with 2 and 4 levels

2 levels (1 bit per level)

```
1 | +---+ +---+
0 | +---+ +---+ +---+
```

4 levels (2 bits per level)

```
11 | +---+
10 | +---+ |
01 | | +---+
00 | +---+ +---+
```

📖 Bits per level

Formula: **bits per level** = $\log_2(L)$, where L is the number of levels.

If the result is not a whole number, **round up** to the next integer.

🔗 Worked Examples

Ex 3.16 / QB Q21: 8 levels $\rightarrow \log_2 8 = 3$ bits per level.

Ex 3.17: 9 levels $\rightarrow \log_2 9 = 3.17$, not realistic; we need a whole number of bits, so **4 bits** per level.

Ex 3.18 / QB Q22: 100 pages per **minute**; assume a page = 24 lines $\times$ 80 characters, each character 8 bits. Bits per page = $24 \times 80 \times 8 = 15,360$. Per minute: $100 \times 15,360 = 1,536,000$ bits. Per second: $1,536,000 / 60 = \mathbf{25,600 \text{ bps} = 25.6 \text{ kbps}}$. *(State the assumption in the exam.)*

Ex 3.19: digitised voice: 4 kHz signal, sampled at twice the highest frequency = 8000 samples/s, 8 bits each $\rightarrow 8000 \times 8 = \mathbf{64,000 \text{ bps} = 64 \text{ kbps}}$.

Ex 3.20: HDTV: 1920×1080 pixels $\times$ 30 screens/s $\times$ 24 bits = **1,492,992,000 bps $\approx$ 1.5 Gbps**; compression reduces this to 20–40 Mbps.

📄 Code snippet: bit-rate numericals

```
import math
def bits_per_level(L):
    return math.ceil(math.log2(L))

print(bits_per_level(8))      # 3
print(bits_per_level(9))      # 4
print(100 * 24 * 80 * 8 / 60) # 25600.0 bps   (QB Q22)
print(8000 * 8)               # 64000 bps    (digitised voice)
print(1920 * 1080 * 30 * 24)  # 1492992000 bps (HDTV)
```

Bit length**📄 Definition (write this in the exam)**

Bit length is the physical distance one bit occupies on the transmission medium as it travels from sender to receiver. **Bit length = propagation speed $\times$ bit duration**

🔗 Worked Example

Assume propagation speed 2×10^8 m/s in a cable and bit rate 1 Mbps (bit duration = 1 μ s). Bit length = $2 \times 10^8 \times 10^{-6} = \mathbf{200 \text{ m}}$. So a 1 Mbps stream stretches each bit over 200 metres of cable.

Baseband transmission and low-pass channels

- **Baseband transmission** means sending a digital signal over a channel **without converting it to an analog signal**. Example: a LAN cable carrying raw voltage pulses.
- A **digital signal is a composite analog signal with infinite bandwidth** (slide 178).
- A **low-pass channel** has a bandwidth that **starts from 0** and goes up to some frequency; baseband transmission needs a low-pass channel.

✅ 30-Second Recap

Bits per level = $\log_2 L$ (round up). Bit rate = total bits / time. Bit length = speed $\times$ bit duration. Baseband = digital signal sent as-is over a low-pass channel.

3.7 Transmission Impairment: Attenuation, Distortion and Noise [★]

ESSENTIAL FOR EXAM]

Exam Insight

Asked as: Question Bank Q23 — "Briefly explain causes of signal impairment / Explain in brief: attenuation, distortion, noise" (3–6 marks). Numericals on dB and SNR (Examples 3.26–3.32) are very likely follow-ups. Slides 180–196.

Simple Analogy

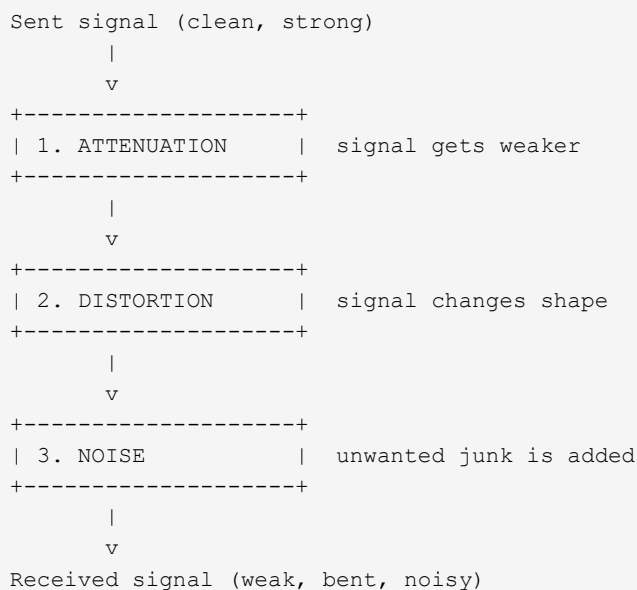
You shout a message across a noisy football ground to a friend.

- **Attenuation:** your voice gets **weaker** with distance (loss of energy).
- **Distortion:** parts of your words arrive at slightly different times, so the message sounds **garbled** (change in shape).
- **Noise:** other people shouting adds **unrelated sounds** (extra unwanted signal).

Definition (write this in the exam)

Transmission impairment: the signal at the end of a medium is not the same as the signal at the beginning because media are imperfect. What is sent is not what is received. The three causes are **attenuation, distortion and noise**.

Three causes of impairment



Cause	What happens	Reason	Remedy
Attenuation	Loss of energy: the signal becomes weaker.	The medium resists the signal, converting some energy into heat.	Amplifiers or repeaters.
Distortion	Change in the shape or form of the signal.	Each frequency component has its own propagation speed, so components arrive with different delays, creating phase differences.	Equalisers; better medium.

Cause	What happens	Reason	Remedy
Noise	Unrelated signal added to the transmitted signal.	Thermal noise (random motion of electrons), induced noise (motors, appliances), crosstalk (one wire affecting another), impulse noise (spikes from power lines).	Shielding, twisted pairs, filters.

Measuring attenuation: the decibel (dB)

Definition (write this in the exam)

dB = $10 \times \log_{10} (P_2 / P_1)$ where P_1 is the input power and P_2 is the output power. A **negative** dB means loss (attenuation); a **positive** dB means gain (amplification).

Power change	dB	Memory trick
× 2 (doubles)	+3 dB	"double = +3"
× 0.5 (halves)	-3 dB	"half = -3"
× 10	+10 dB	Each ×10 adds 10 dB
× 100	+20 dB	
× 1000	+30 dB	
× 0.1	-10 dB	

Worked Examples (slides 183–188)

Ex 3.26: power halves, $P_2 = \frac{1}{2}P_1$. $\text{dB} = 10 \log_{10}(0.5) = -3 \text{ dB}$ (a loss of 3 dB equals losing half the power).

Ex 3.27: amplifier increases power 10×. $\text{dB} = 10 \log_{10}(10) = +10 \text{ dB}$.

Ex 3.28: dB values **add** along a chain. Example chain: -3 dB (cable) + 7 dB (amplifier) - 3 dB (cable) = +1 dB overall, so you do not need to multiply power ratios.

Ex 3.29: dBm = $10 \log_{10}(P_{\text{mW}})$. For dBm = -30: $\log_{10} P = -3$, so **P = $10^{-3} \text{ mW} = 0.001 \text{ mW} = 1 \mu\text{W}$** .

Ex 3.30: cable loses 0.3 dB/km; the input is 2 mW; length 5 km. Loss = $5 \times (-0.3) = -1.5 \text{ dB}$. $-1.5 = 10 \log_{10}(P_2/2) \rightarrow P_2 = 2 \times 10^{(-0.15)} \approx 1.41 \text{ mW}$.

Signal-to-Noise Ratio (SNR)

Definition (write this in the exam)

SNR = average signal power / average noise power, and **SNR_dB** = $10 \times \log_{10} (\text{SNR})$. A **high SNR** means the signal is much stronger than the noise (good). A **low SNR** means noise dominates (bad).

High vs low SNR

HIGH SNR (good link)

signal: /\ /\ /\
noise: ~ (tiny wiggles)
receiver can separate them

LOW SNR (bad link)

signal: /\ /\ /\
noise: ~~~~~~ (big wiggles)
signal is buried in noise

🔗 Ex 3.31 and Ex 3.32

Ex 3.31: signal power = 10 mW, noise power = 1 μ W = 0.001 mW. $\text{SNR} = 10 / 0.001 = \mathbf{10,000}$. $\text{SNR}_{\text{dB}} = 10 \log_{10}(10,000) = \mathbf{40 \text{ dB}}$.

Ex 3.32 (noiseless channel): noise power = 0, so $\text{SNR} = \infty$ and $\text{SNR}_{\text{dB}} = \infty$. This ideal can **never** be achieved in real life.

📄 Code snippet: dB and SNR

```
import math
def db(p_out, p_in):      return 10 * math.log10(p_out / p_in)
def snr_db(ps, pn):      return 10 * math.log10(ps / pn)

print(round(db(0.5, 1), 2))      # -3.01   (Ex 3.26)
print(round(db(10, 1), 2))      # 10.0    (Ex 3.27)
print(2 * 10 ** (-1.5 / 10))    # 1.4158 mW (Ex 3.30)
print(snr_db(10, 0.001))       # 40.0    (Ex 3.31)
```

⚠️ Common Mistake

Using **ln** instead of **log₁₀**, or forgetting the **10×** factor.

Mixing units: convert μ W and mW to the **same unit** before dividing.

Forgetting that attenuation is **negative dB** and amplification is **positive dB**.

☑️ 30-Second Recap

Attenuation = weaker, distortion = reshaped, noise = extra junk. $\text{dB} = 10 \log_{10}(P_2/P_1)$: halve = -3 dB , $\times 10 = +10 \text{ dB}$.

$\text{SNR} = \text{signal power} / \text{noise power}$; $\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$. Higher SNR is better.

UNIT 4 — Switching

What this unit is about: how a large network moves data from any source to any destination without wiring every device to every other. Chapter 8, slides 197–231. Question Bank Q55–58 ask: list the switching techniques, then explain circuit, message and packet switching with advantages and disadvantages.

4.1 Why Switching? Taxonomy of Switched Networks [★ ESSENTIAL FOR EXAM]

Exam Insight

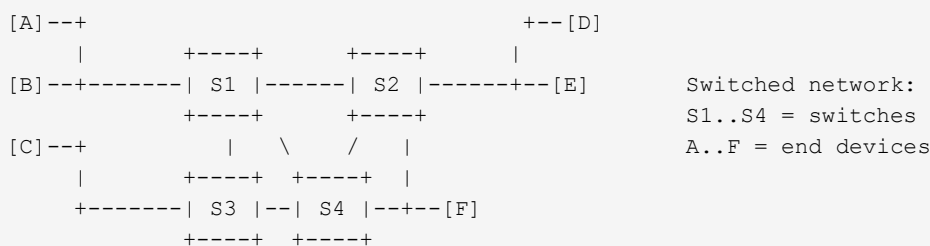
Asked as: Question Bank Q55 — "List switching techniques and explain them briefly." Slides 198–203.

Simple Analogy

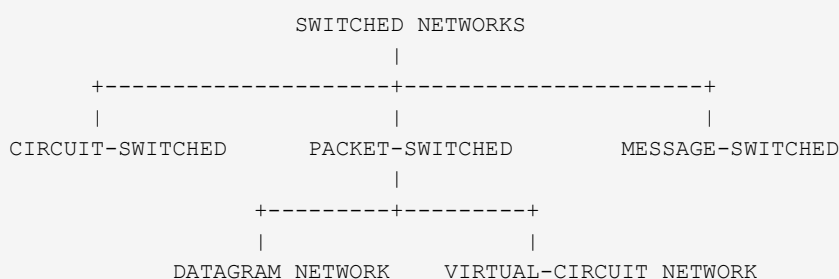
Connecting 1000 houses with a **private road between every pair** (mesh) is impossible; one **central roundabout** (star) overloads; one **long road** (bus) jams. Real cities use **intersections and junctions** that connect roads temporarily and direct traffic. **Switches are those junctions.**

- For a very large network, mesh and star topologies are impractical, and a bus cannot cope once distance and device count exceed the limits of the medium. **Switching is the better solution.**
- A **switch** is a hardware and/or software device that creates **temporary connections** between two or more devices linked to the switch but not to each other.
- A **switched network** is a series of interconnected nodes (switches). Some connect to end devices; others only route traffic.

A switched network



Taxonomy of switching techniques



Exam Tip

Today the tendency is to **combine** datagram and virtual-circuit ideas: the first packet is routed by datagram addressing, then a virtual circuit is created for the rest of the packets from the same source to the same destination.

4.2 Circuit Switching [★ ESSENTIAL FOR EXAM]

Exam Insight

Asked as: Question Bank Q56 — "Explain circuit switching with its advantages and disadvantages." Slides 204–213.

Simple Analogy

A **traditional telephone call** or a **reserved railway berth**. Before you speak, the exchange reserves a private line from you to your friend. Whether you talk or stay silent, that line is yours until you hang up; nobody else can use it.

Definition (write this in the exam)

A **circuit-switched network** consists of a set of switches connected by physical links. A connection between two stations is a **dedicated path** made of one or more links; each connection uses **one dedicated channel on each link**. Each link is divided into **n channels** using **FDM or TDM**.

Three phases

Circuit switching: setup, data transfer, teardown

Caller A	Switches	Callee B
---- 1. SETUP request ----> ----> ----> ----->		reserve a channel on
<----- ACK / connected - <----- <----- <-----		every link of the path
===== 2. DATA TRANSFER (dedicated path) =====>		resources stay reserved
<===== no per-packet routing =====		even during silence
---- 3. TEARDOWN -----> ----> ----> ----->		release the channels

- **Setup phase:** resources (channels, switch buffers/ports) are **reserved** end to end before any data is sent.
- **Data transfer phase:** data flows over the dedicated path with **no addressing needed**; resources stay dedicated for the **entire duration**.
- **Teardown phase:** the connection is released and the resources become free again.

Pseudocode: three phases

```
# Pseudocode: circuit switching
setup(A, B):
    path = find_path(A, B)
    for link in path:
        if link.has_free_channel(): reserve(link.channel)
        else: return "BUSY"          # call blocked
    return "CONNECTED"

transfer(A, B): send data over reserved channels (no routing per frame)
teardown(A, B): for link in path: release(link.channel)
```

Efficiency and delay

- **Efficiency (low):** the reserved channel is wasted during silence (pauses in a phone call, idle computer connections). Resources cannot be shared.
- **Delay (small during transfer):** there is a **setup delay** and a **teardown delay**, but once connected data flows with only **propagation delay** and negligible switch delay; there is **no queuing** at switches.
- Total delay = setup time + data transfer time (transmission + propagation) + teardown time.

 Delay in a circuit-switched network

```

time ->
|  setup delay  |      data transfer delay      | teardown |
|=====>|=====>|=====>|
  (request+ack travel across all switches)  (message + propagation)

```

 Worked Example (Example 8.1): 8 telephones

Eight telephones in a small area, each using a **4 kHz voice channel**. Each link uses **FDM** to carry a maximum of **two** voice channels, so each link has bandwidth $2 \times 4 \text{ kHz} = 8 \text{ kHz}$.

Connections: telephone 1↔7, 2↔5, 3↔8, 4↔6. The switch controls the connections and they change as new calls are made.

Example 8.2: two remote offices of a company connected by a leased **T-1 line**, with two 4×8 switches (4 inputs, 8 outputs). Four outputs loop back for same-office calls; four go to the other office.

 Circuit-switched network of Example 8.1 (simplified)

```

tel 1 --\                               /-- tel 5
tel 2 ---\                             /--- tel 6
tel 3 ----+---[ SWITCH ]=====+----- tel 7    "====" link = 2 FDM
tel 4 ---/      1-7, 2-5, 3-8, 4-6      \---- tel 8    voice channels (8 kHz)

```

Advantages	Disadvantages
• Dedicated path: guaranteed bandwidth, steady data rate. • No delay variation (jitter) once connected: ideal for real-time voice. • No queuing delay at switches. • Data arrives in order over one path. • Simple data transfer: no routing or addressing per packet.	• Setup time before any data can be sent. • Poor resource utilisation: idle reserved channels cannot be used by others. • Expensive: capacity is reserved even if unused. • Inflexible: a break or failure drops the call. • Blocking: a new call is refused if no free channel.

 Real-World Example

The **traditional telephone network** (switching at the physical layer uses the circuit-switching approach).

 30-Second Recap

Circuit switching = reserve a dedicated channel (FDM/TDM) for the whole call: setup → transfer → teardown. Great for continuous voice; wasteful for bursty data.

4.3 Datagram Networks (Packet Switching) [**ESSENTIAL FOR EXAM**]

 Exam Insight

Asked as: Question Bank Q55, Q57, Q58 ("Explain packet switching with its advantages and disadvantages"). Slides 214–221.

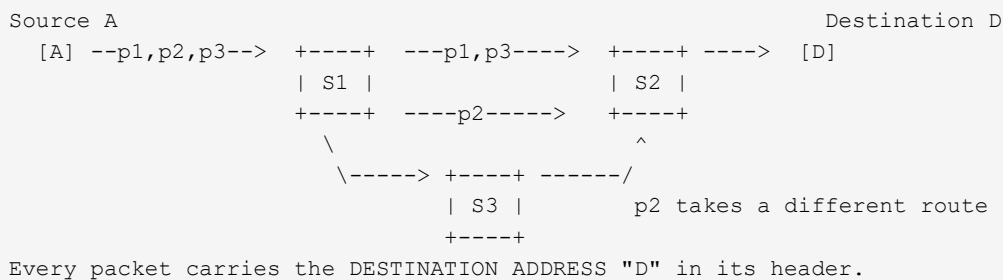
 Simple Analogy

Mailing a **long book through the post one page at a time**. Each page is put in its own envelope with the destination address and posted separately. Pages may travel by different routes, arrive in a different order, or even one may get lost. The receiver reassembles them.

Definition (write this in the exam)

In **packet switching** the message is divided into **packets** of fixed or variable size. There is **no resource reservation**; resources are **allocated on demand**. In a **datagram network** each packet is treated **independently**, as if it were the only packet.

- Each switch (router) uses a **routing table based on the destination address**.
- The **destination address in the packet header stays the same** during the entire journey.
- Packets of the same message may take **different routes**, arrive **out of order**, with **different delays**, or be **lost or dropped**. Reordering is done at the destination.
- **No setup or teardown phase**.

Datagram network: packets take independent routes

Destination address	Output port
1232	1
4150	2
9130	3

Example routing table of one switch: a packet with destination 4150 is forwarded out of port 2. Addresses and ports are illustrative.

Pseudocode: datagram forwarding

```

# Pseudocode: forwarding in a datagram switch
routing_table = {1232: 1, 4150: 2, 9130: 3}      # destination -> output port

on packet arrival(p):
    port = routing_table[p.destination]           # look at destination address only
    queue p on output port 'port'                 # store and forward
    # no connection state is kept between packets

```

- **Efficiency (high):** no reserved resources; lines are shared on demand.
- **Delay:** no setup or teardown; each packet has its own delay = transmission + propagation + **queuing** + processing at each switch. Delays can differ between packets.

Advantages	Disadvantages
• No setup time: data can be sent immediately. • Efficient use of bandwidth: links shared by many users. • Robust: packets can bypass a failed link or switch. • Flexible and cheap for bursty data. • Fits the Internet.	• Packets may arrive out of order or be lost; reordering needed. • Variable delay (jitter) because of queuing. • Overhead: every packet carries a full destination address. • Congestion can occur at switches. • No guaranteed bandwidth or quality of service.

 Real-World Example

The Internet: switching at the network layer uses the datagram approach (IP packets).

4.4 Virtual-Circuit Networks [★ ESSENTIAL FOR EXAM]

 Exam Insight

Slides 222–231. Often asked as "differentiate between datagram and virtual-circuit", or inside the four-way comparison (4.6).

 Simple Analogy

Booking a **fixed bus route**. You first book the route (setup). After that, all passengers from your group travel along the **same fixed route**, but they share the bus with other people, so each person may reach the destination at a slightly different time.

 Definition (write this in the exam)

A **virtual-circuit (VC) network** is a **cross between a circuit-switched network and a datagram network**. It has three phases like circuit switching (setup, data transfer, teardown) but data is sent as **packets** that share the links, like a datagram network.

- **Addressing:** a **global address** is used only during **setup**. During data transfer each packet carries a short **virtual-circuit identifier (VCI)** instead of the full destination address.
- The **VCI is local** to each link and **changes at each switch**. Each switch has a **table** mapping $*(\text{incoming port, incoming VCI})^* \rightarrow *(outgoing \text{ port, outgoing VCI})^*$.
- **All packets of the same source–destination pair travel the same path**, but with on-demand resource allocation they may reach the destination with **different delays**.
- **Setup phase:** a **setup request** travels from the source to the destination, each switch makes a table entry; the destination returns a **setup acknowledgment** that carries the VCIs back.
- **Teardown phase:** the source sends a teardown request and the switches delete their table entries.

 Virtual-circuit identifiers change hop by hop

```
[A] --(VCI 14)--> [S1] --(VCI 22)--> [S2] --(VCI 71)--> [B]
```

```
Switch S1 table:  in (port 1, VCI 14)  ->  out (port 3, VCI 22)
```

```
Switch S2 table:  in (port 2, VCI 22)  ->  out (port 1, VCI 71)
```

Packets carry only a short VCI (14, then 22, then 71), NOT the full address.

Every packet of this circuit follows exactly A -> S1 -> S2 -> B.

 Phases in a virtual-circuit network

Source	S1	S2	Destination	
--SETUP req-->	--SETUP req-->	--SETUP req-->		1. setup request
<--SETUP ack--	<--SETUP ack--	<--SETUP ack--		2. setup ack (VCIs)
==== data packets with VCI, same path =====>				3. data transfer
--TEARDOWN--->	--TEARDOWN--->	--TEARDOWN--->		4. teardown

- **Efficiency:** resource reservation can be done in the **setup phase** (as in circuit switching) or **on demand** (as in datagram), so efficiency lies between the two.
- **Delay:** setup and teardown delay plus per-packet transmission/propagation/queuing delay; packets of the same circuit are not re-routed and **stay in order**.

 Real-World Example

Switched WANs (e.g. Frame Relay, ATM) implement switching at the **data link layer** using virtual circuits (slide 231).

4.5 Message Switching [ESSENTIAL FOR EXAM]

 Exam Insight

Asked as: Question Bank Q57 — "Explain message switching with its advantages and disadvantages." Slide 203.

 Simple Analogy

Sending a **parcel through a chain of post offices**. Each office takes the **entire parcel**, stores it in the warehouse, and forwards it to the next office when the road is free. The parcel is never cut into pieces.

 Definition (write this in the exam)

In **message switching**, each switch **stores the whole message** and then **forwards it** to the next switch (store-and-forward). There is no dedicated path and no packetising.

 Message switching: store and forward whole messages

```
[Sender] --message--> [Switch 1] --message--> [Switch 2] --message--> [Receiver]
                        store, then                store, then
                        forward                    forward
```

Advantages	Disadvantages
• No setup phase and no dedicated path. • Efficient channel use: links are shared and used only when messages move. • Sender need not wait for the receiver to be online. • Handles priority messages and can broadcast to many.	• Large storage needed at each switch for whole messages. • Long delays: a whole message is received before being forwarded. • Unsuitable for real-time or interactive traffic. • A large message blocks the link for others.

 Real-World Example

Electronic mail (e-mail) is the common surviving example: mail servers store the entire message and forward it. Message switching is not used at lower layers today.

4.6 Comparison of Switching Techniques [ESSENTIAL FOR EXAM]

 Exam Insight

The most efficient revision table for Unit 4: it covers "differentiate" questions on any pair of switching techniques.

Feature	Circuit switching	Datagram (packet)	Virtual circuit	Message switching
Setup phase	Yes	No	Yes	No
Resource reservation	Yes, for the whole call	No, on demand	Setup or on demand	No

Feature	Circuit switching	Datagram (packet)	Virtual circuit	Message switching
Unit of data	Continuous stream	Packets (independent)	Packets (same path)	Whole message
Path	One dedicated path	Each packet may differ	One fixed path per circuit	Hop-by-hop store-and-forward
Addressing	During setup only	Full destination address in every packet	Global address at setup; short VCI after	Full address in message
Packet order	In order	May be out of order	In order	In order
Delay	Setup delay; small afterwards, no queuing	Variable; queuing at each switch	Setup delay + variable	Long (whole message per hop)
Link efficiency	Low	High	Medium to high	Medium
Suitable for	Voice calls	Internet / bursty data	Switched WANs	E-mail
Layer / example	Physical layer: telephone network	Network layer: Internet	Data link layer: switched WAN	Application: e-mail

⚠ Common Mistake

Saying virtual circuit **reserves a physical wire**. It does not; the path is "virtual" because packets still share links.

Forgetting that in virtual-circuit networks the **VCI changes at every switch**.

Saying datagram packets always arrive in order. They **do not**.

☑ 30-Second Recap

Circuit = reserve whole path. Datagram = independent packets, routing table by destination. Virtual circuit = setup a path, then packets with VCI. Message = store whole message and forward.

REVISION TOOLKIT — Formula Sheet, Practice Questions and Checklist

Use this section in the last 24 hours. Everything here points back to a numbered section of the notes.

R.1 One-Page Formula Sheet

Topic	Formula / rule	Section
Mesh links (full-duplex)	$n(n - 1) / 2$	1.8
Mesh links (simplex)	$n(n - 1)$	
Mesh ports per device	$n - 1$	1.8
Bit stuffing	Insert 0 after every five consecutive 1s; flag = 01111110	2.2
Byte stuffing	Insert ESC before every FLAG or ESC in the data	2.2
Stop-and-Wait ARQ	Sequence numbers 0/1; ACK = next expected frame; windows 1 and 1	2.5
Go-Back-N windows	Sender $2^m - 1$, receiver 1	2.7
Selective Repeat windows	Sender 2^{m-1} , receiver 2^{m-1}	2.8
m from max sequence number	$2^m = \text{max sequence number} + 1$	2.10
Bandwidth-delay product	bandwidth $\times$ round-trip time	2.5
Link utilisation	(window $\times$ frame bits) / BDP, maximum 100%	2.10
Frequency and period	$f = 1 / T$	3.2
Wavelength	$\lambda = v / f = v \times T$	3.2
Phase	1 cycle = $360^\circ = 2\pi$ rad	3.2
Bandwidth of a signal	$B = f_{\text{high}} - f_{\text{low}}$	3.5
Non-periodic range from middle	$f_{\text{low}} = \text{mid} - B/2$; $f_{\text{high}} = \text{mid} + B/2$	3.5
Bits per signal level	$\log_2 L$ (round up)	3.6
Bit rate	total bits / time	3.6
Bit length	propagation speed $\times$ bit duration	3.6
Decibel	$\text{dB} = 10 \log_{10} (P_2 / P_1)$	3.7
dBm	$\text{dBm} = 10 \log_{10} (P \text{ in mW})$	3.7
SNR	$\text{SNR} = P_{\text{signal}} / P_{\text{noise}}$; $\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$	3.7

R.2 Practice Questions with Answer Skeletons

Cover the right column, write your own answer under exam conditions, then compare. Questions are ordered by priority.

#	Question	Marks	Answer skeleton (where to find it)
1	🔥 Why does Selective-Repeat ARQ perform better than Go-Back-N ARQ? Justify with a diagram and critical comments.	6–8	Intro $\rightarrow$ side-by-side lost-frame diagram $\rightarrow$ 8-row table $\rightarrow$ critical comments + verdict. (2.9)

#	Question	Marks	Answer skeleton (where to find it)
2	🔥 Explain Selective Repeat ARQ and show how it gives error control and flow control, with an example.	8	Definition → windows $2^m - 1$ → buffering and NAK → lost-frame timeline → pros/cons. (2.8)
3	🔥 Explain Go-Back-N ARQ in detail.	6–8	Definition → window $2^m - 1 / 1$ → cumulative ACK → lost-frame timeline (frames 3–6 resent) → pseudocode idea → pros/cons. (2.7)
4	🔥 Explain Stop-and-Wait ARQ with its four cases.	6–8	Features → normal / lost frame / lost ACK / delayed ACK diagrams → 5% utilisation example. (2.5)
5	🔥 Illustrate bit stuffing with an example.	6	Definition → rule → worked example (stuff and unstuff) → why needed. (2.2E)
6	🔥 Why is framing required at the data link layer? Explain any three header fields.	6	3 reasons → frame diagram → destination, source, control/sequence fields. (2.2A–B)
7	🔥 Differentiate bit stuffing and byte stuffing.	4–6	7-row table + one mini-example of each. (2.2F)
8	🔥 Differentiate Star, Bus, Mesh and Ring topologies with diagrams.	8	4 diagrams → master comparison table → advantages/disadvantages in one line each. (1.8)
9	🔥 How many cables and ports are needed for 5 devices in a mesh?	4	10 cables, 4 ports per device, 20 ports in total. Show the formula. (1.8)
10	🔥 Explain the key characteristics of PAN, LAN, MAN and WAN.	8	Range · ownership · speed · technology · example for each + comparison table. (1.9)
11	🔥 Draw the frequency domain: non-periodic signal, $B = 100$ kHz, middle 150 kHz, peak 10 V, extremes 0.	3	$f_l = 100$ kHz, $f_h = 200$ kHz; smooth curve peaking at 10 V at 150 kHz. (3.5)
12	🔥 Periodic signal with components 100, 300, 500, 700, 900 Hz: find bandwidth, draw spectrum.	2–4	$B = 800$ Hz; five 10 V spikes. (3.5)
13	★ Distinguish unicast, multicast and broadcast.	6	Three small diagrams + comparison table. (1.1)
14	★ Define point-to-point and multipoint connections.	2–3	Definition + diagram + example each. (1.7)
15	★ A digital signal has eight levels. How many bits per level?	1–2	$\log_2 8 = 3$ bits. (3.6)
16	★ Required bit rate to download 100 pages per minute?	1–3	$24 \times 80 \times 8 \times 100 / 60 = 25,600$ bps (state the page-size assumption). (3.6)
17	★ Explain attenuation, distortion and noise.	3–6	Definition → cause → effect → remedy for each + figure. (3.7)
18	★ A signal loses half its power. Find the attenuation in dB.	2–3	$10 \log_{10}(0.5) = -3$ dB. (3.7)
19	★ Signal 10 mW, noise 1 μ W: find SNR and SNR _{dB} .	2–3	SNR = 10,000; SNR _{dB} = 40 dB. (3.7)
20	★ GBN with maximum sequence number 15: sender and receiver window sizes?	1–2	$m = 4$ → sender 15, receiver 1. (2.10)
21	★ SR with maximum sequence number 15: window sizes?	1–2	$m = 4$ → sender 8, receiver 8. (2.10)
22	★ Explain circuit switching with advantages and disadvantages.	5–8	Definition → three phases diagram → delay/efficiency → pros/cons table → telephone example. (4.2)

#	Question	Marks	Answer skeleton (where to find it)
23	★ Explain packet switching (datagram) with advantages and disadvantages.	5–8	Packets → routing table → diagram → pros/cons → Internet example. (4.3)
24	★ Explain message switching with advantages and disadvantages.	5–8	Store-and-forward diagram → pros/cons → e-mail example. (4.5)
25	★ Compare circuit switching, datagram and virtual-circuit networks.	6–8	Comparison table in 4.6 plus a one-line example for each.

R.3 Diagrams to Draw from Memory (10-minute self-test)

1. The four topologies (mesh with 4 nodes, star, bus with taps and terminators, ring with arrow).
2. PAN → LAN → MAN → WAN size ladder.
3. Frame structure: FLAG | header | payload | trailer | FLAG.
4. Bit stuffing example: 0110111111011111100 → 0110111111010111110100.
5. Stop-and-Wait ARQ: lost-frame and lost-ACK timelines.
6. Go-Back-N lost-frame timeline (frames 2, 3, 4 resent).
7. Selective Repeat lost-frame timeline (only frame 2 resent, buffered 3 and 4 delivered).
8. Sliding window with SF, SL and the window sliding after a cumulative ACK.
9. Frequency spectrum of a periodic (spikes) and a non-periodic (curve) signal.
10. Circuit-switching three phases, and a virtual-circuit table with changing VCI.

R.4 Last-Minute Reminders

- **Every answer needs a diagram or an example.** The papers instruct "draw neat sketches" and "assume suitable data and mention the same".
- For **comparison questions**, a table scores faster than paragraphs: aim for 6–8 rows.
- For **numericals**, always write: formula → substitution → answer with units.
- **Time split for a 6-mark answer:** 1 minute thinking, 2 minutes diagram, 4 minutes writing.
- Highest-value topics in order: **ARQ (2.5–2.9) → framing (2.2) → topologies (1.8) → network categories (1.9) → bandwidth numericals (3.5).**

☒ Final word

Understand the analogy, draw the figure, solve one example, then write it in your own words. Good luck with your sessional!